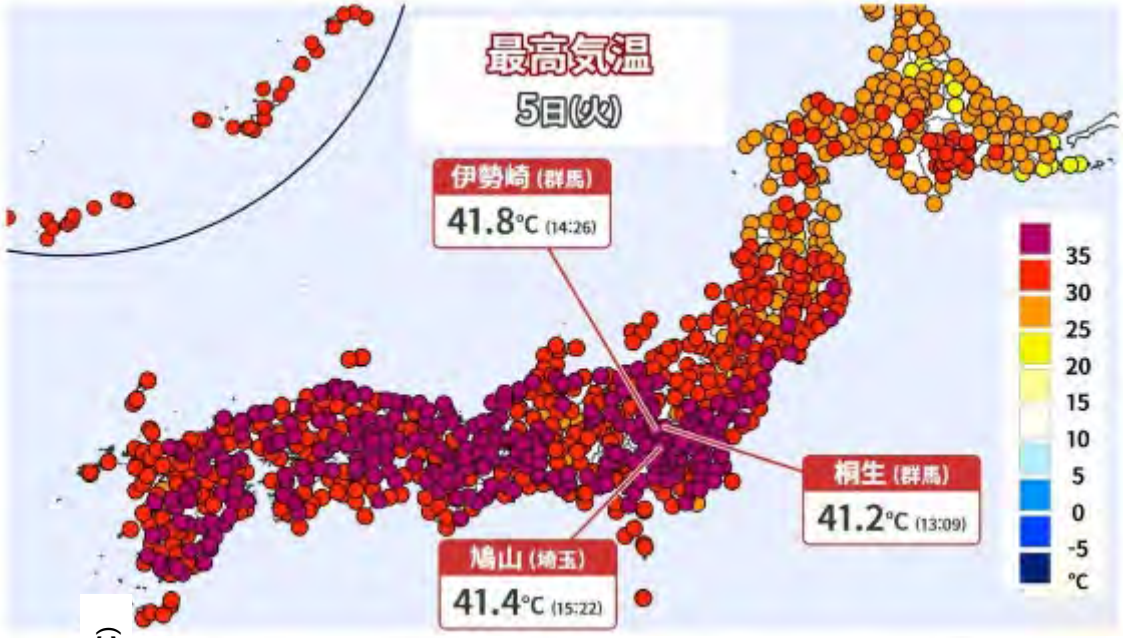
Recent Heat-Related Illness Prevention Measures in Japan



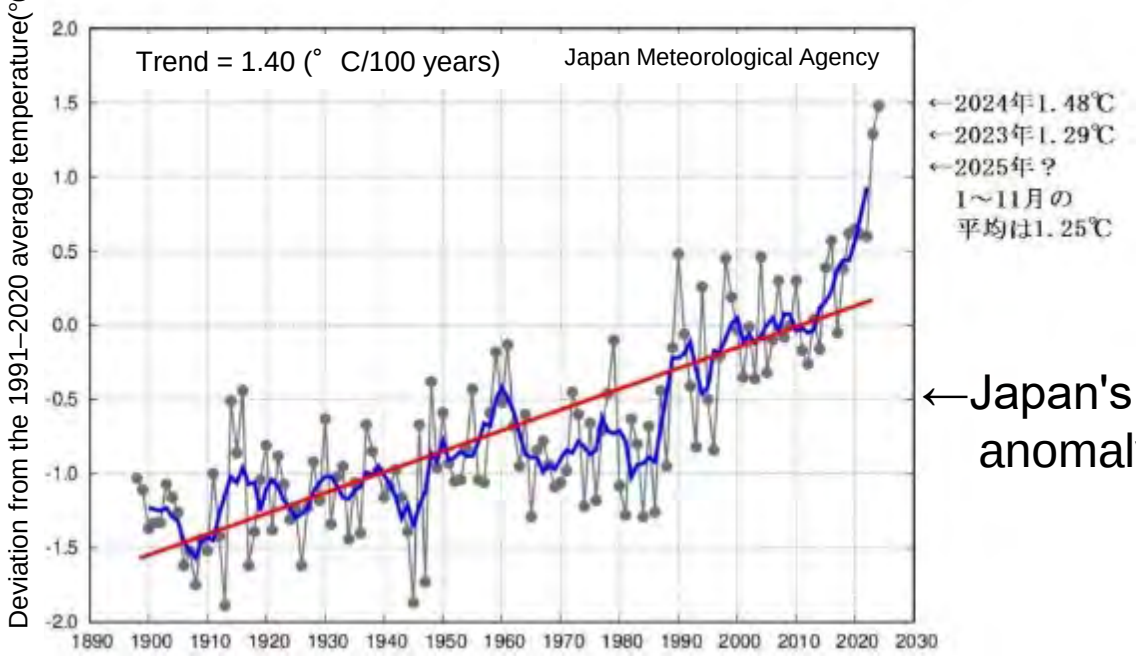




The average global temperature in 2025 was the hottest in history.



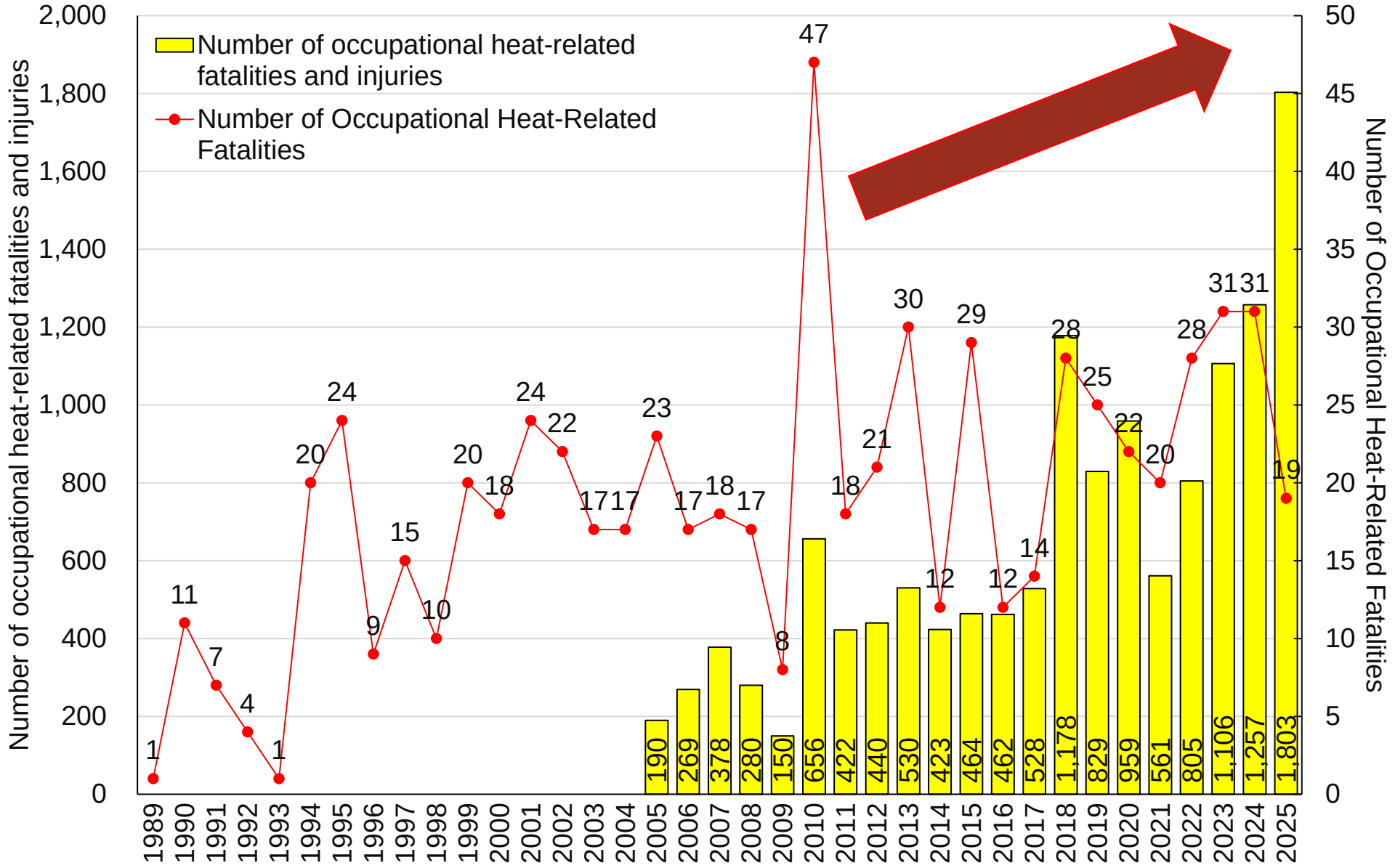




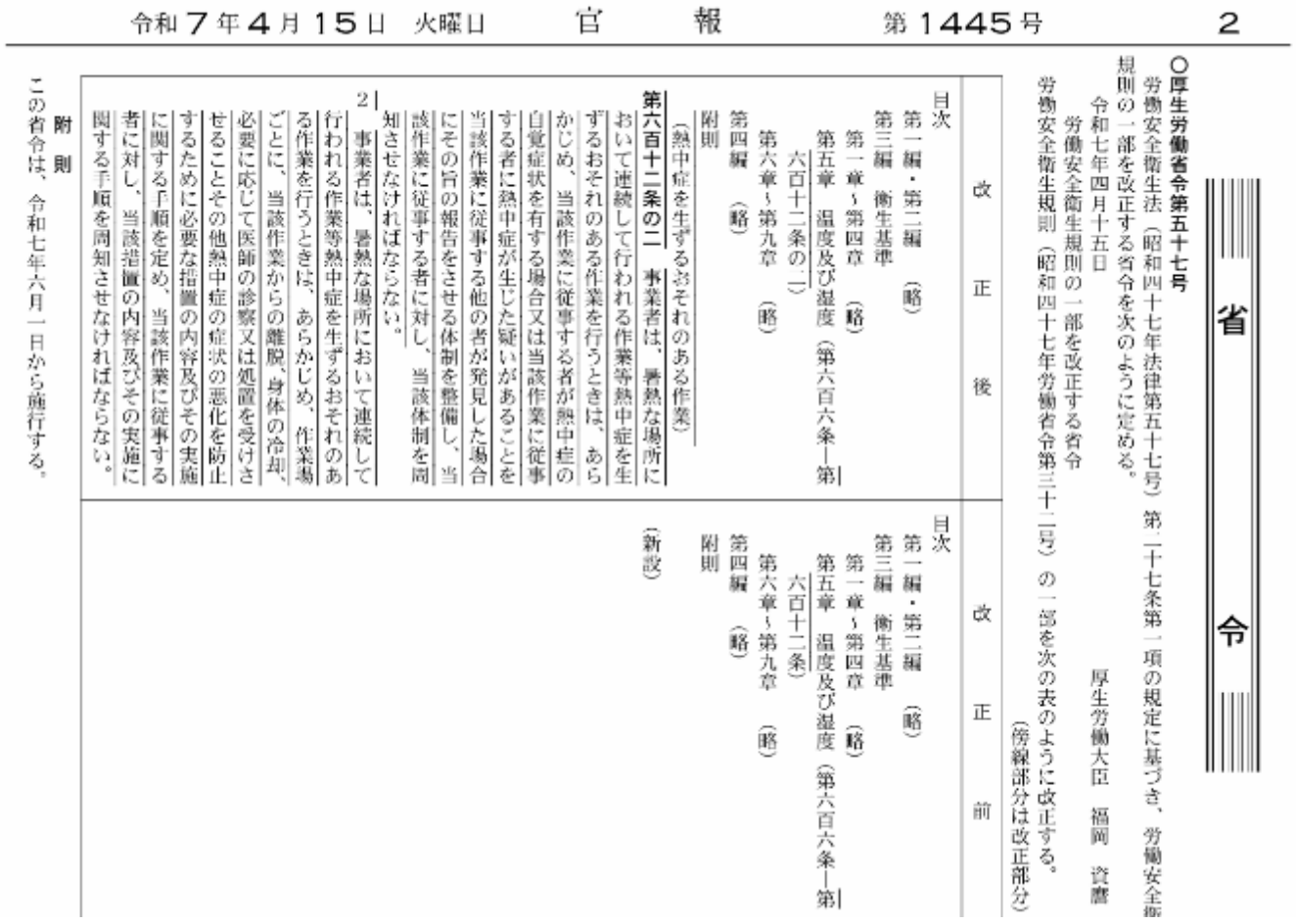
Updated highest temperature in Japan (2025/08/05)

←Japan's annual average temperature anomaly

Occupational Heat-Related Casualties in Japan



POLICIES FOR THE PREVENTION OF HEAT ILLNESS IN JAPAN



令和７年４月15日　火曜日　　官　　報　　第1445号　　2

省　令

○厚生労働省令第五十七号

労働安全衛生法（昭和四十七年法律第五十七号）第二十七条第一項の規定に基づき、労働安全衛生規則の一部を改正する省令を次のように定める。

令和七年四月十五日

厚生労働大臣　福岡　資麿

労働安全衛生規則の一部を改正する省令

労働安全衛生規則（昭和四十七年労働省令第三十二号）の一部を次の表のように改正する。

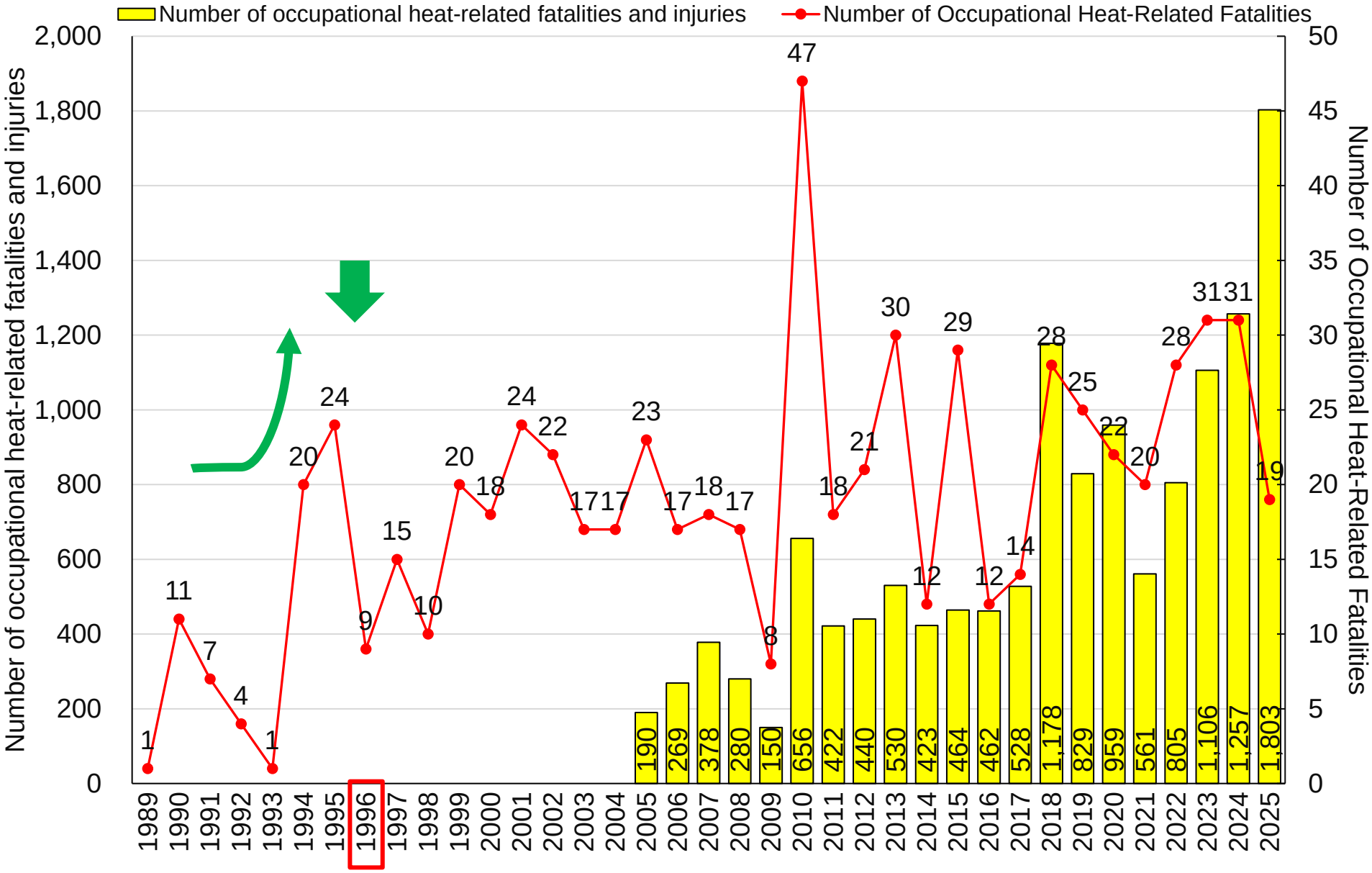
（傍線部分は改正部分）

改正後	改正前
目次	目次
第一編・第二編　（略）	第一編・第二編　（略）
第三編　衛生基準	第三編　衛生基準
第一章～第四章　（略）	第一章～第四章　（略）
第五章　温度及び湿度（第六百六条―第六百十二条の二）	第五章　温度及び湿度（第六百六条―第六百十二条）
第六章～第九章　（略）	第六章～第九章　（略）
第四編　（略）	第四編　（略）
附則	附則
（熱中症を生ずるおそれのある作業）	
第六百十二条の二　事業者は、暑熱な場所において連続して行われる作業等熱中症を生ずるおそれのある作業を行うときは、あらかじめ、当該作業に従事する者が熱中症の自覚症状を有する場合又は当該作業に従事する者に熱中症が生じた疑いがあることを当該作業に従事する他の者が発見した場合にその旨の報告をさせる体制を整備し、当該作業に従事する者に対し、当該体制を周知させなければならない。	（新設）
2　事業者は、暑熱な場所において連続して行われる作業等熱中症を生ずるおそれのある作業を行うときは、あらかじめ、作業場ごとに、当該作業からの離脱、身体の冷却、必要に応じて医師の診察又は処置を受けさせることその他熱中症の症状の悪化を防止するために必要な措置の内容及びその実施に関する手順を定め、当該作業に従事する者に対し、当該措置の内容及びその実施に関する手順を周知させなければならない。	

附　則

この省令は、令和七年六月一日から施行する。

“Prevention of Heat Illness” (Notice No. 329, May 21, 1996)



Occupational Heat-Related Casualties in Japan

“Prevention of Heat Illness”

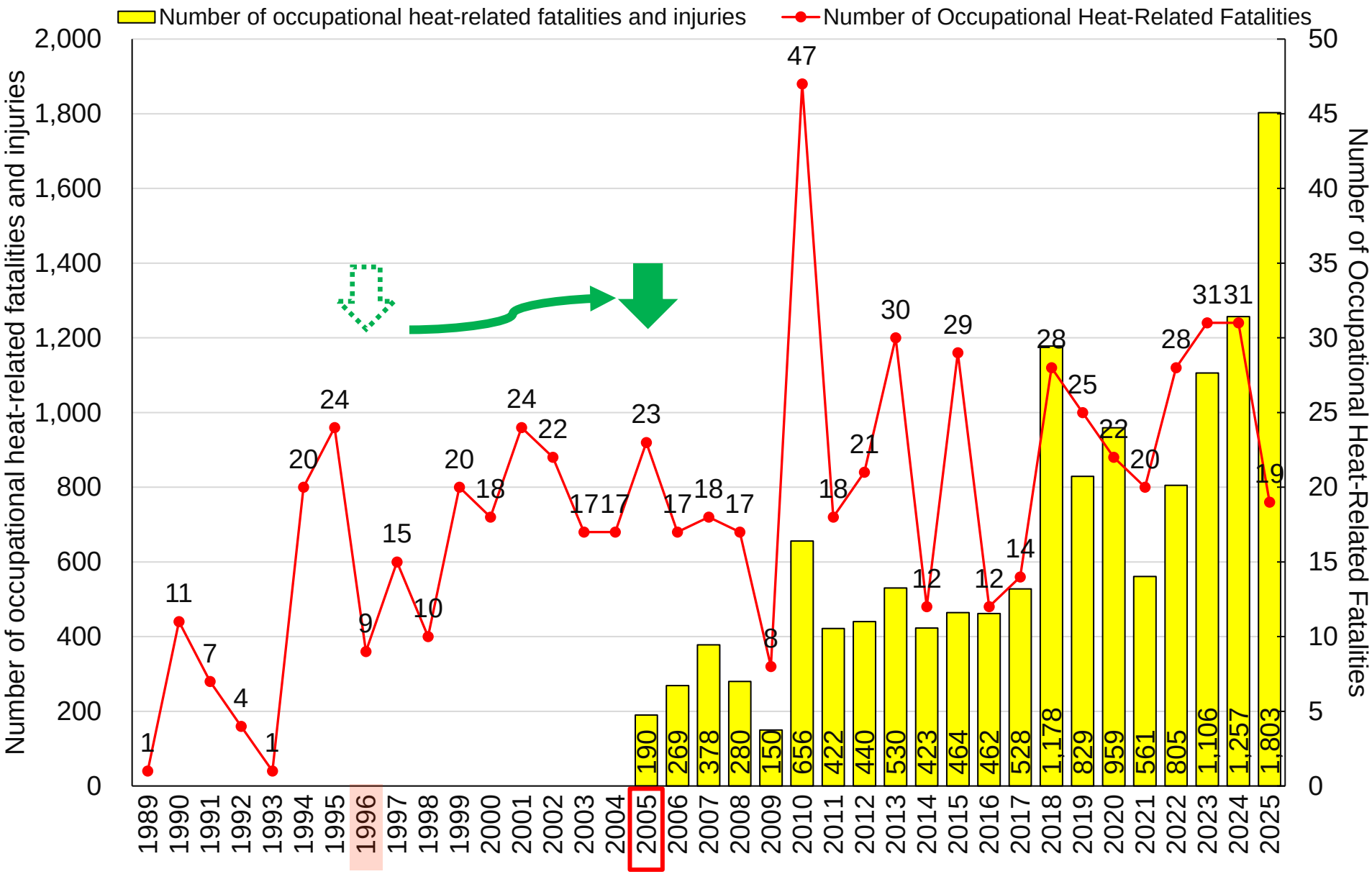
(Notice No. 329, May 21, 1996)

First occupational heat illness prevention guidance applicable to all industries,including workplaces without heat sources.

1. Work environment management
2. Work management
3. Health management
4. Occupational Health Education
5. Emergency response

“Guidance on the Use of WBGT for Heat Illness Prevention”

(Notice No. 0729001, July 29, 2005)



Occupational Heat-Related Casualties in Japan

“Guidance on the Use of WBGT for Heat Illness Prevention”

(Notice No. 0729001, July 29, 2005)

The measurement of WBGT at the workplaces and the utilization of the value for appropriate countermeasures.

WBGT thermal stress index reference value corresponding to each condition

Level	Workload	WBGT(℃)			
		Aclimatized		Unaclimatized	
0	Rest	33		32	
1	Low metabolic rate	30		29	
2	Moderate metabolic rate	28		26	
		No air flow	Air flow present	No air flow	Air flow present
3	High metabolic rate	25	26	22	23
4	Extremely high metabolic rate	23	25	18	20

WBGT（Wet Bulb Globe temperature）

- An apparent temperature index used to estimate the effects of temperature, humidity, wind speed, and solar radiation on humans.

Outdoor (with solar radiation)

:WBGT = 0.7 T_w + 0.2 T_g + 0.1 T_d

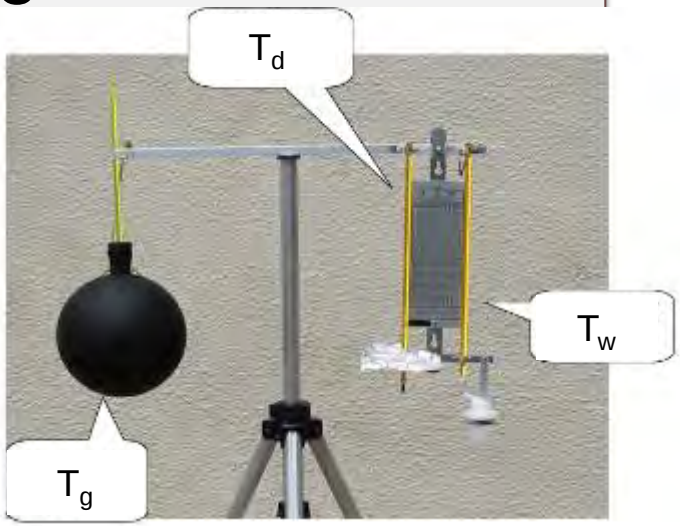
Indoor or outdoor (without solar radiation)

:WBGT = 0.7 T_w + 0.3 T_g

T_w: Natural wet-bulb temperature

T_g: Globe temperature

T_d: Dry-bulb temperature (ambient air temperature)

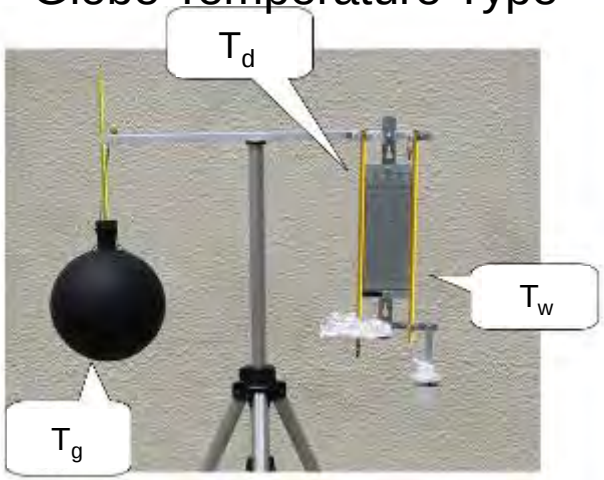


WBGT vs Heat Index

	WBGT	Heat Index (HI)
Purpose	Workplace heat stress assessment	Public heat-risk communication
Input variables	Temperature, humidity, solar radiation	Temperature and humidity
Output	Occupational heat stress level	Apparent temperature felt by humans
Users	Occupational health professionals, employers	General public, workers
Main application	Work-rest schedules, heat stress management	Heat warnings and risk alerts
Measurement requirements	Requires a WBGT meter	Can be calculated from weather data

ISO 7243 / JIS Z8504

Natural Wet-Bulb and Globe Temperature Type







JIS B7922

Humidity Sensor and Globe Temperature Type









Without Globe Thermometer





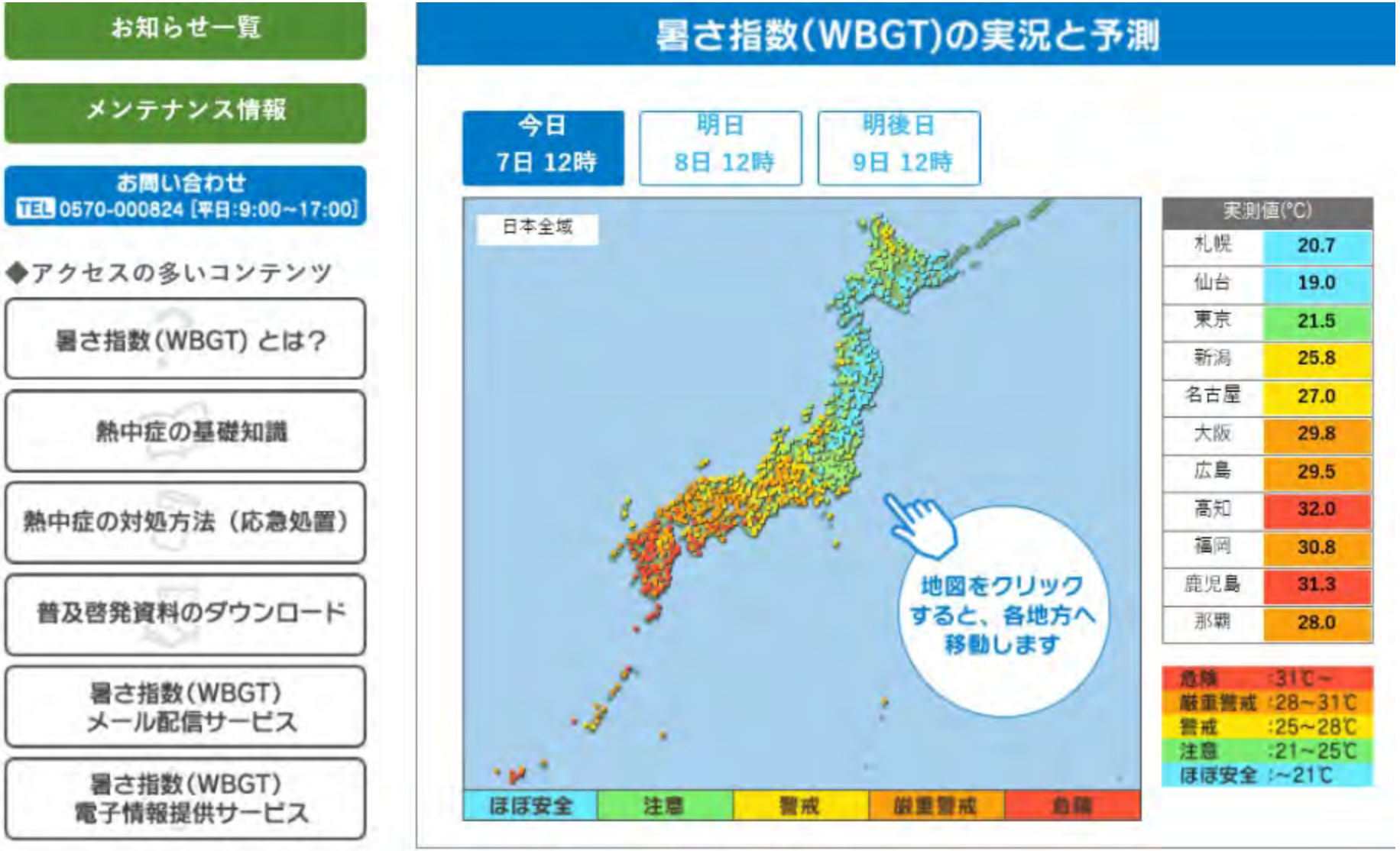




Expensive — Cheap

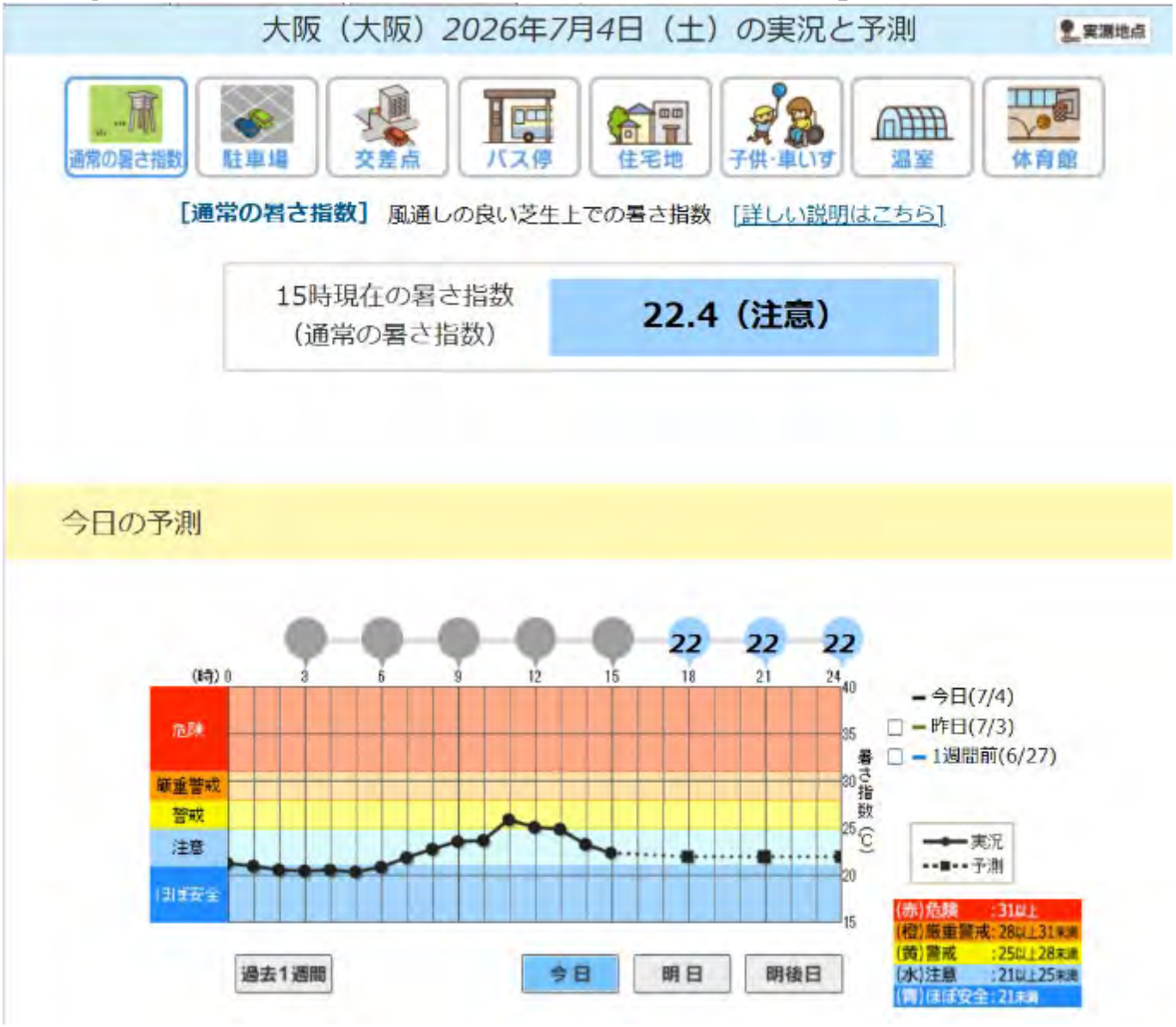
WBGT Measuring instrument

Service of WBGT report and forecast on the web
Ministry of the Environment, Japan



http://www.wbgt.env.go.jp/

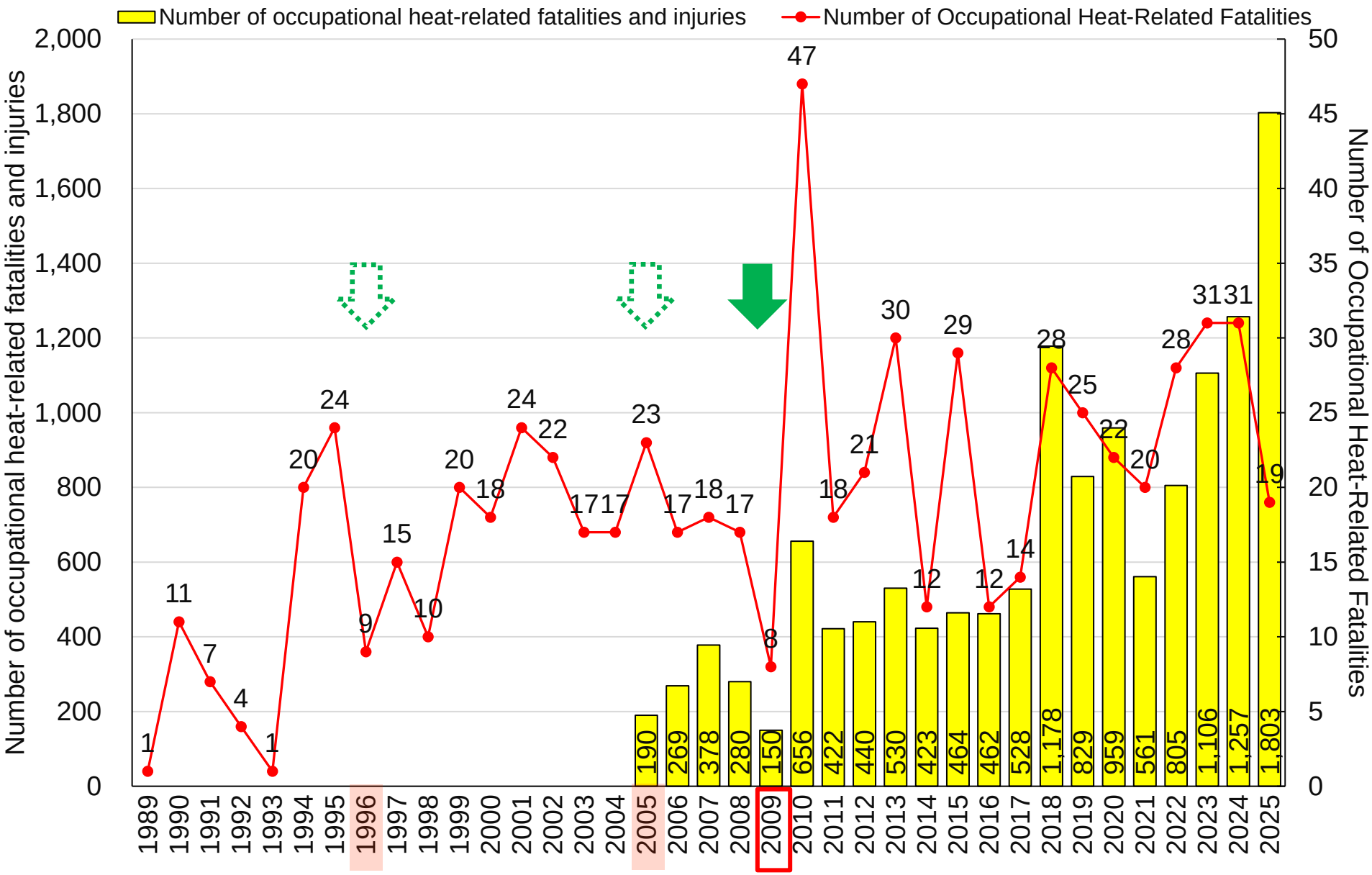
Service of WBGT report and forecast on the web
Ministry of the Environment, Japan



http://www.wbgt.env.go.jp/

“Guidelines for the Prevention of Occupational Heat Illness”

(Notice No. 0619001, June 19, 2009)



Occupational Heat-Related Casualties in Japan

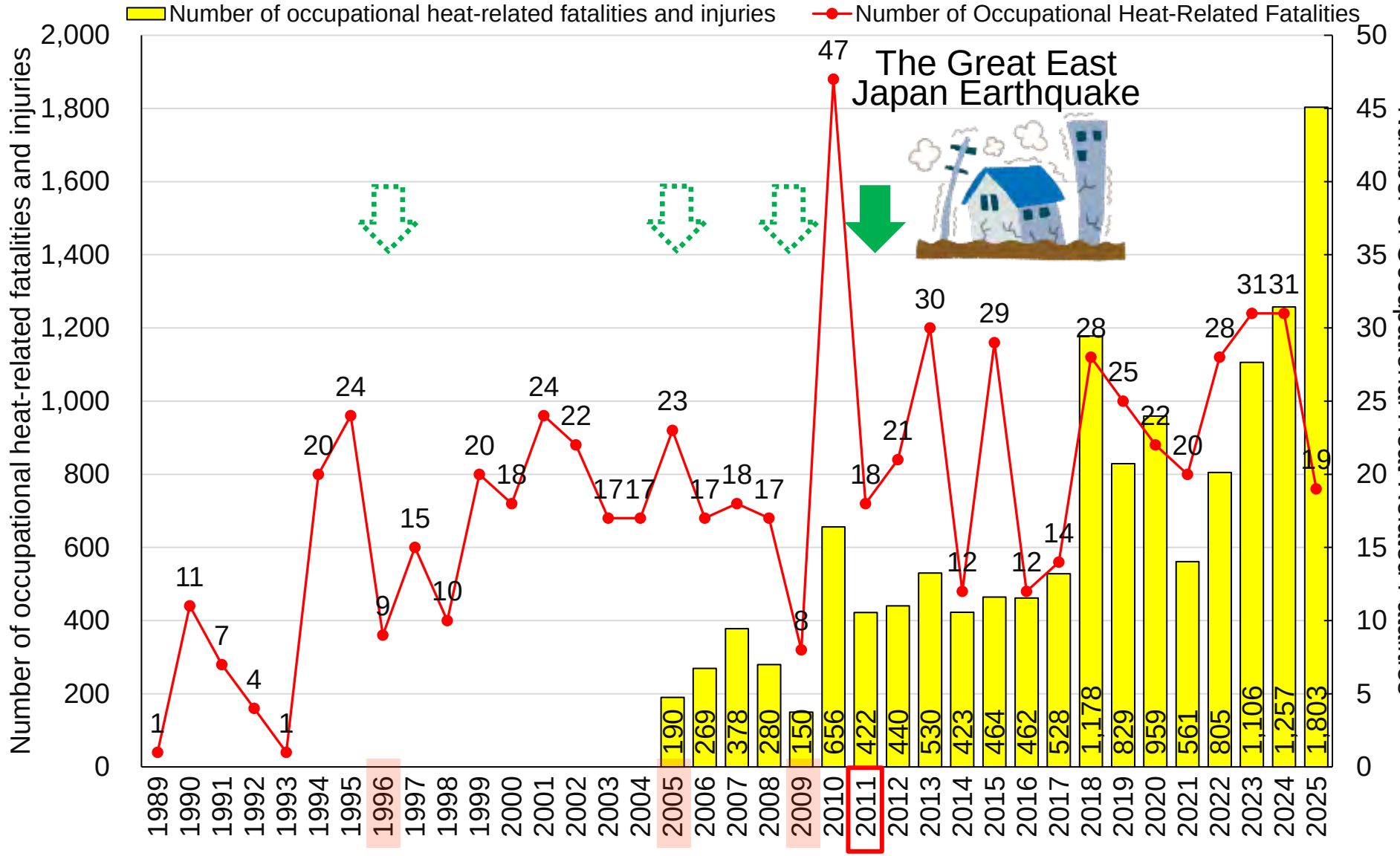
“Guidelines for the Prevention of Occupational Heat Illness”

(Notice No. 0619001, June 19, 2009)

- Integrated heat illness prevention framework
  - **WBGT-based assessment**
  - **Work management**
    - ✓Planned Heat Acclimatization
    - ✓Appropriate Water and Salt Intake
  - **Health management**
    - ✓Consideration for workers with underlying medical conditions
    - ✓Management of daily health conditions
  - **Emergency response**

“Measures Following Summer Electricity-Saving Policies”

(Notice No. 0520006, May 20, 2011)



Occupational Heat-Related Casualties in Japan

“Handling of Indoor Office Temperature and Related Conditions in Response to Summer Electricity Supply and Demand Measures”

(Notice No. 0520006, May 20, 2011)

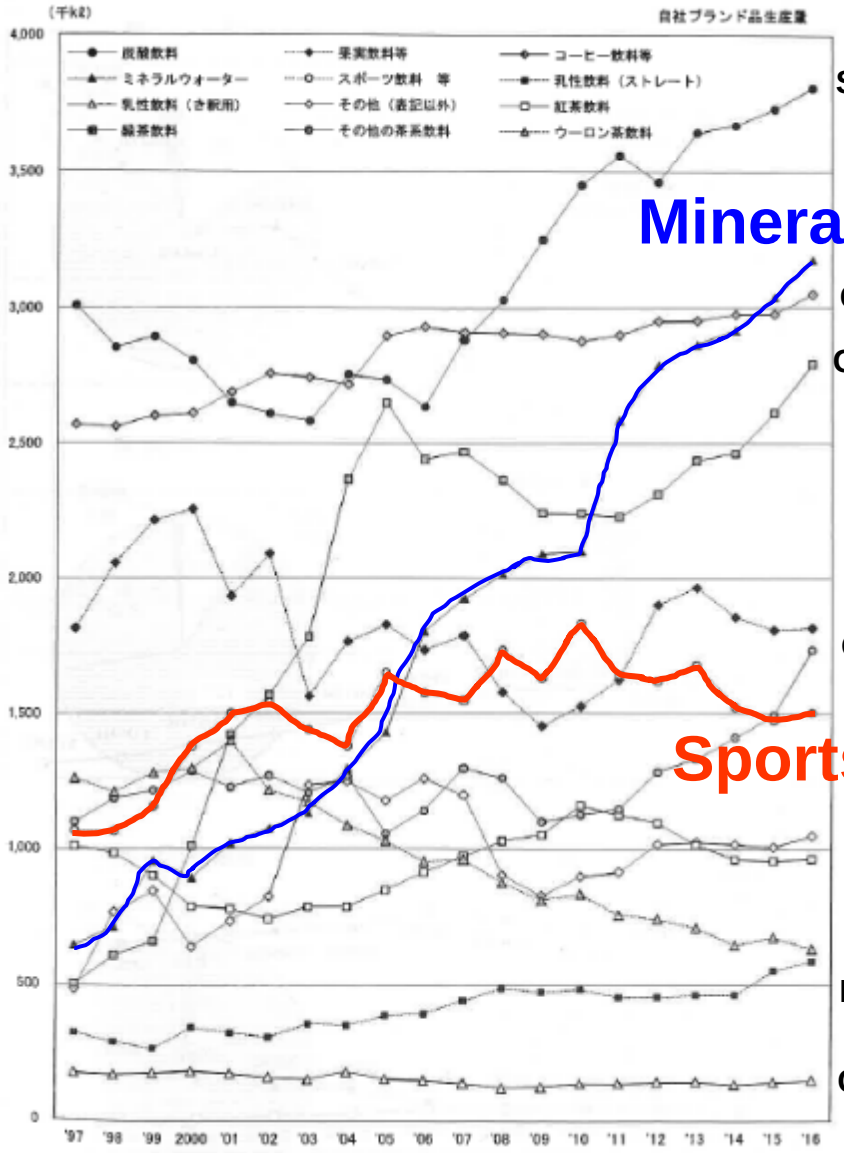
- Promotion of national energy-saving measures amid challenges in operating nuclear power plants
  - Recommended office temperature not exceeding 28°C
  - Additional precautions required when temperatures exceed 29°C
  - Prevention of heat illness and excessive physical or mental burden through **non-electric** measures
    - ✓ Sports Drinks and Oral Rehydration Solutions
    - ✓ Salt Supplementation (Tablets or Candy)
    - ✓ Body-Cooling Workwear
    - ✓ Quick-Drying, Cooling-Fabric Clothing



Sports drink



Oral rehydration



Soda drink

Mineral Water

Coffee

Green tea

Fruits juice

Other tea

Sports drink

Others

Black tea

Oolong tea

Milk drink

Calpico,etc.

“Soft Drink Production”,
（Japan Soft Drink Manufacturers Association, 2017）

Sports drink and Oral rehydration in Japan

Salt Tablets and Salt Candy

: Help Replenish Sodium Lost Through Sweating



Recommended to take these with water

Workers wearing protective devices

- **Garments developed to lower workers' body temperature**



➢ *Water cooling type*



➢ *Compressed air type*



➢ *Ice pack type*



➢ *Fan-Assisted Workwear*

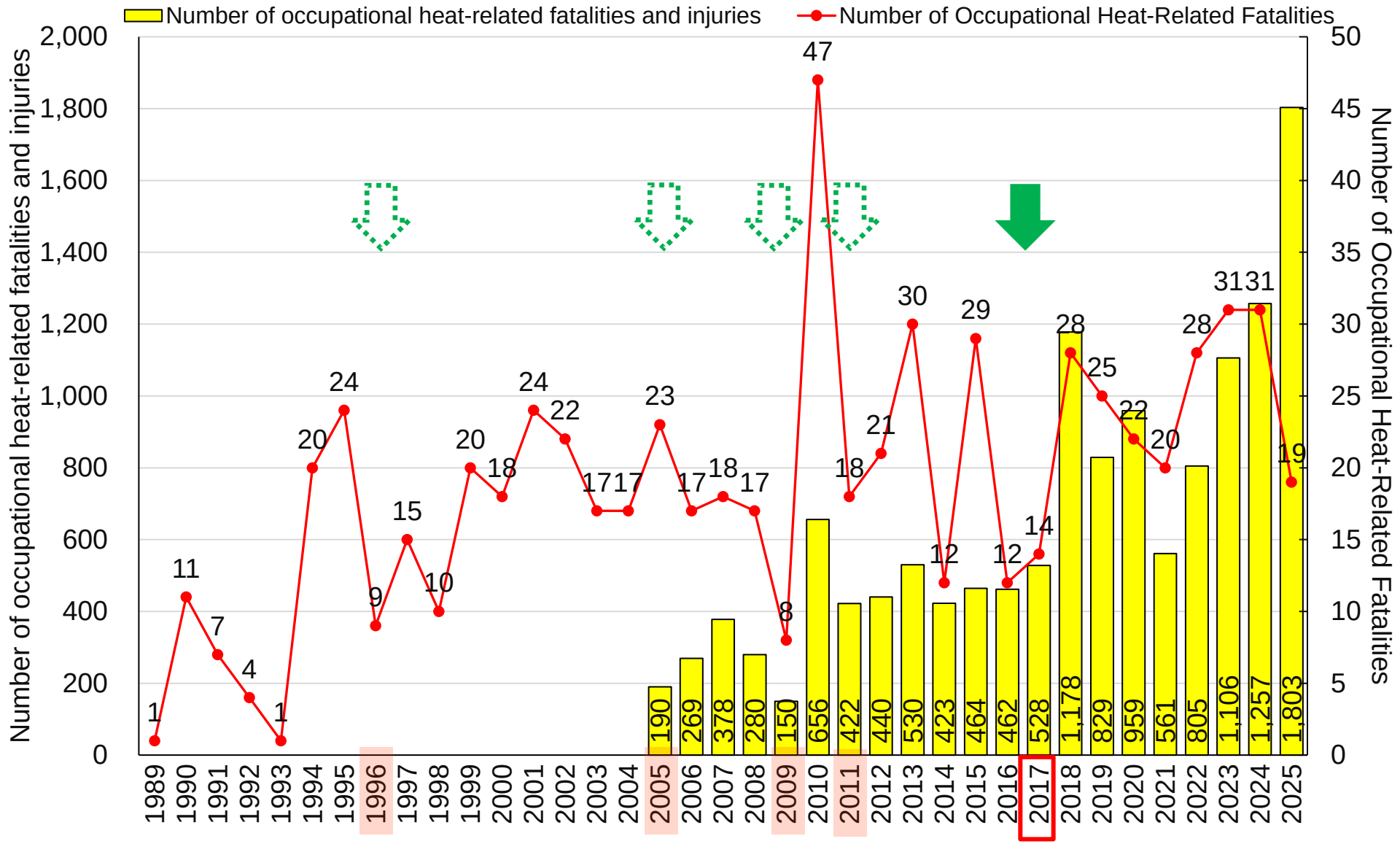


Cool work clothes used at nuclear power plants after the Great East Japan Earthquake, wearing with protective clothing from radiation



“STOP! Heat Illness Cool Work Campaign”

Annual notice issued before summer since 2017



Occupational Heat-Related Casualties in Japan

“STOP! Heat Illness Cool Work Campaign”

STOP!

中暑
清凉工作活动



近年来，因工作期间中暑
每年约有30人死亡，
1,000人以上请假超过四天。



活动期间

4月	5月	6月	7月	8月	9月
准备			重点实施		

准备期 4月 应采取的措施

确认并 ☑ 检查是否已切实实施各项措施。

- ☐ **建立职业卫生管理体系**
  选任每个工作场所的预防中暑管理人员，并建立明确的预防中暑责任体系。
- ☐ **监测热指数（WBGT）的准备**
  采用符合JIS标准的设备测量热指数（WBGT），并定期检查。
- ☐ **制定作业计划**
  制定作业计划，包括根据热指数（WBGT）确保休息时间、作业暂停相关事项。
- ☐ **探讨设备配备**
  探讨安装简易遮阳棚、通风设备、或空调、喷淋系统，以降低热指数（WBGT）。
- ☐ **探讨确保休息区域**
  探讨设立具备空调房的休息区域或凉爽的休息区域。
- ☐ **探讨服装**
  准备透气性和排汗性良好的服装，也考虑穿着具有通过送风或送水对身体进行降温的功能的服装。
- ☐ **实施教育培训**
  对管理人员和劳动者进行教育培训。
  视频及教育视频　在线学习（e-learning）
- ☐ **事先确认发生紧急情况时的应对措施**
  确认发生紧急情况时的应对措施（发生异常情况时的联络体系及应对流程），并通知相关人员。

厚生労働省・都道府県労働局・労働基準監督署　厚生劳动省、都道府县劳动局、劳动基准监督署　2025.2

活动期间 5月～9月 应采取的措施



环境省预防中暑信息网站

STEP 1 **监测并评估热指数（WBGT）**
使用符合JIS标准的热指数测定仪随时监测WBGT。
参考具有地区代表性的一般热指数（环境省提供）也是有效手段。

STEP 2 **根据测量的热指数（WBGT）贯彻以下措施。**

- ☐ **降低热指数**
  实施准备期制定的设备相关措施。
- ☐ **配备休息区**
  设立在准备期探讨的休息区域。
- ☐ **服装**
  穿着准备期探讨的服装。
- ☐ **缩短作业时间**
  基于作业计划实施根据热指数（WBGT）进行休息或暂停作业。
- ☐ **预冷**
  在作业开始前和休息期间降低核心体温。
- ☐ **摄取水分及盐分**
  定期摄取水分及盐分（考虑随身携带水等）。
- ☐ **热适应措施**
  为适应高温，作业时间的调整应为7天以上。尤其特别关注新入职员工和休假归来的员工。
- ☐ **基于健康检查结果的应对措施**
  对患有以下疾病的人员，根据医生等的意见予以适当限制：①糖尿病 ②高血压 ③心脏病、④肾功能障碍 ⑤精神及神经类疾病 ⑥大面积皮肤病 ⑦感冒 ⑧腹泻
- ☐ **日常健康管理**
  提醒当天未摄入早餐、睡眠不足、前一天大量饮酒可能会引发中暑，并在作业开始前进行确认。
- ☐ **确认作业中的劳动者健康状况**
  频繁进行巡查并与劳动者交流，采取“伙伴制度”（Buddy System），指导劳动者彼此关注健康状况。
- ☐ **异常情况的应对措施**
  切实传达事先制定的联络体系和应对流程。
  如果察觉本人或周围的人员有异常，应立即按照事先制定的联络体系和应对流程妥善处理。
  必须立即停止作业，用水湿润全身并进行送风来降温。
  如果症状未缓解，应立即将患者送往医院（根据症状拨打急救电话）

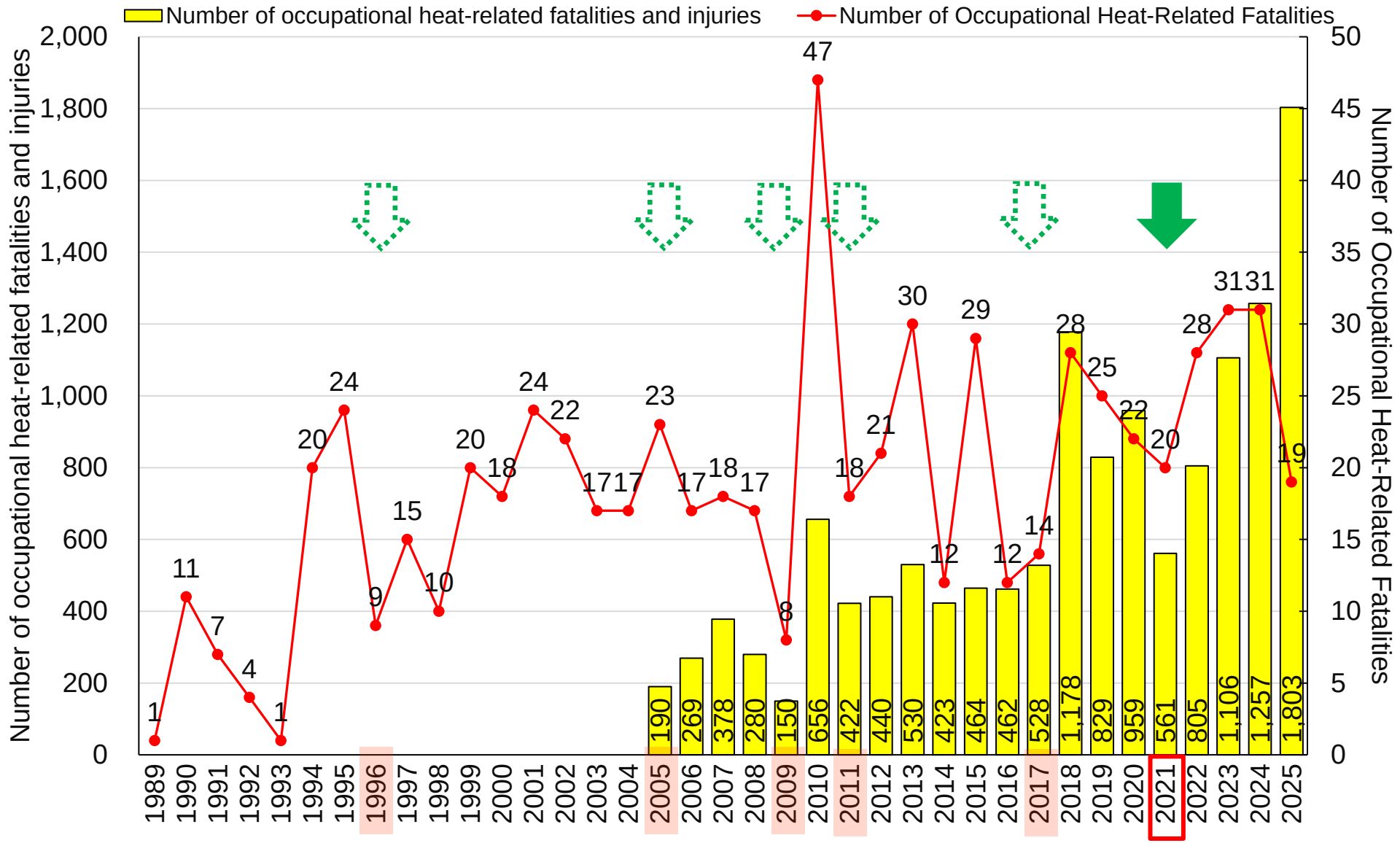
重点实施期间 7月 应采取的措施



- ☐ 重新确认降低热指数的效果，并根据需要追加措施。
- ☐ 贯彻执行根据热指数中断作业等措施。
- ☐ 切实确保劳动者积极补充水分和盐分。
- ☐ 贯彻确认劳动者在作业开始前的健康状况，增加巡查频率。
- ☐ 实施包含中暑风险升高的相关教育。
- ☐ 如果发现身体不适人员有任何异常，应立即拨打急救电话。

“Basic Guidelines for Heat Illness Prevention”

(Notice No. 0420-3, April 20, 2021)



Occupational Heat-Related Casualties in Japan

“Basic Guidelines for Heat Illness Prevention”

(Notice No. 0420-3, April 20, 2021)

- Background of the 2021 Revision
  - Persistent occurrence of occupational heat illness and fatalities
  - Revision of JIS Z 8504 after approximately 20 years
  - Update of WBGT reference values and clothing adjustment factors

Partial Revision in 2021 & 2025

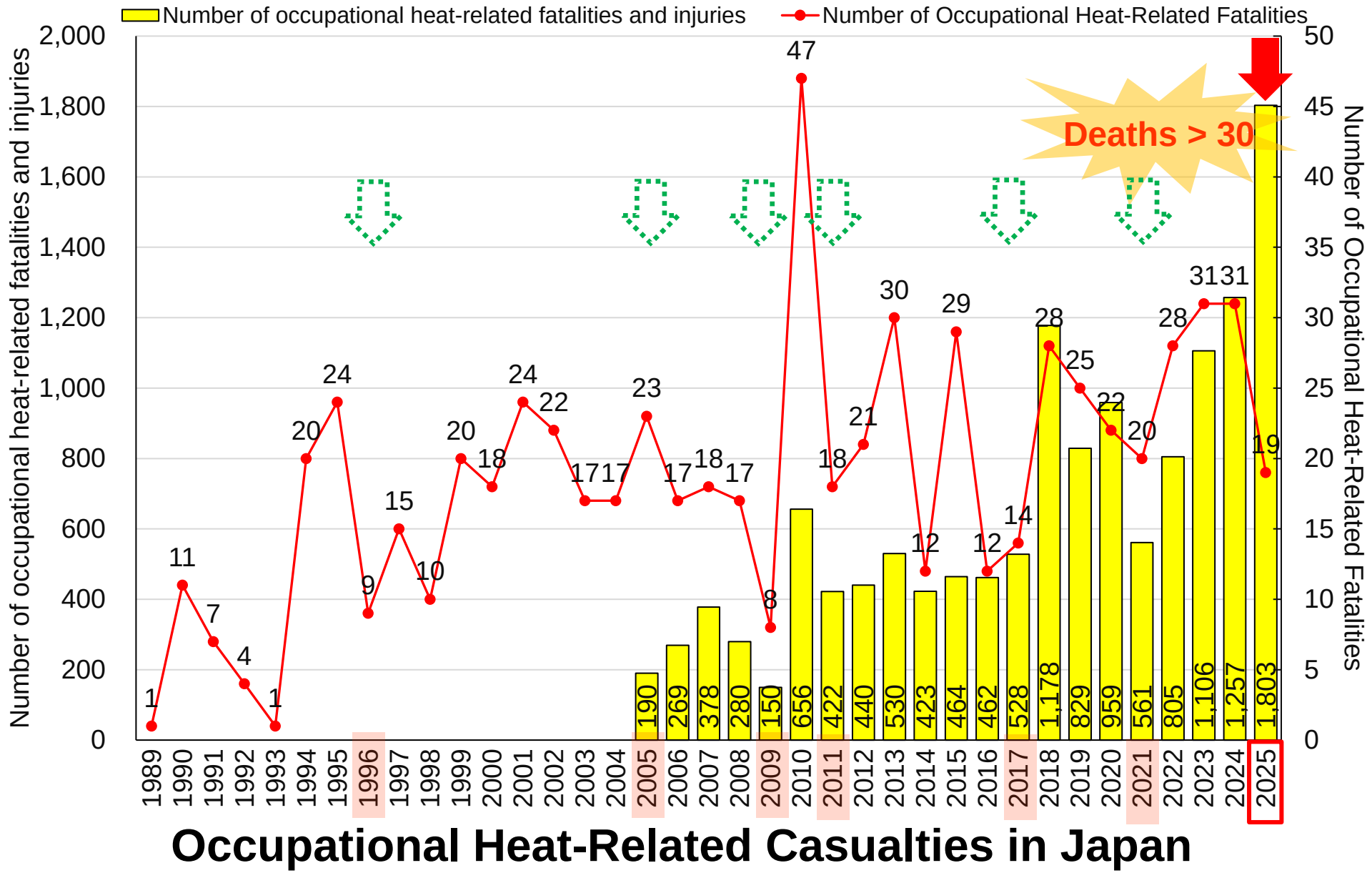
Strengthened Heatstroke Measures under the Amended Climate Change Adaptation Act (fully enforced in April 2024)

	Heatstroke Alert	Heatstroke Special Alert
Purpose	To raise awareness of dangerous heat conditions and encourage preventive actions	To call for the highest level of alert and action when unprecedented and widespread heat is expected to cause a surge in heat-related emergencies and potentially disrupt healthcare services
Issuance Criteria	Forecasted daily maximum WBGT of **33˚C or higher** at any observation point within the target area	Forecasted WBGT of **35˚C or higher** at all heat index observation points within a prefecture, or similarly extreme conditions
Coverage	Forecast regions (58 regional blocks nationwide)	Prefecture
Timing	Around 5:00 PM on the previous day and 5:00 AM on the day of issuance	Around 2:00 PM on the previous day
Recommended Actions	Individual preventive measures, such as air-conditioner use, hydration, and adjustment of outdoor activities	Individual preventive measures plus strong calls for action by families, communities, employers, schools, and event organizers, including cancellation, postponement, or modification of activities

Source: Ministry of the Environment, Japan, https://www.wbgt.env.go.jp/

“Amendment to the Industrial Safety and Health Regulations”

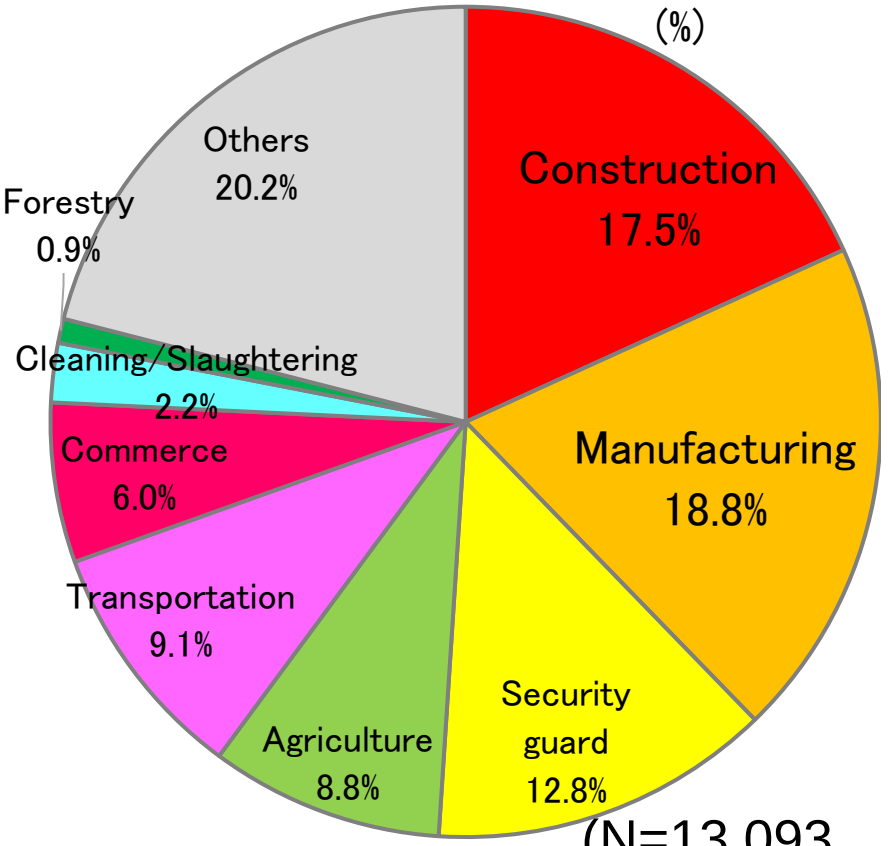
(Promulgated on April 15, 2025; Enforced on June 1, 2025)



Occupational Heat-Related Casualties in Japan

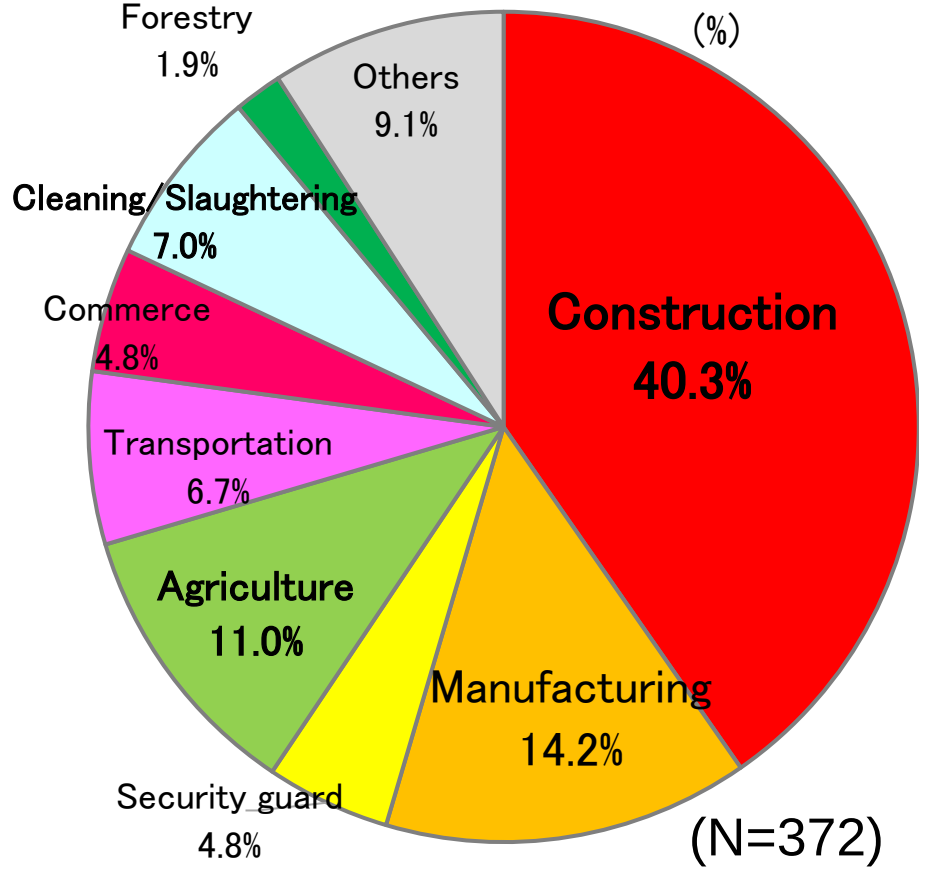
Proportion of heat-related illnesses among workers in Japan, by industry (2004–2025)

Casualties
(injuries and fatalities)



(N=13,093
)

Fatalities



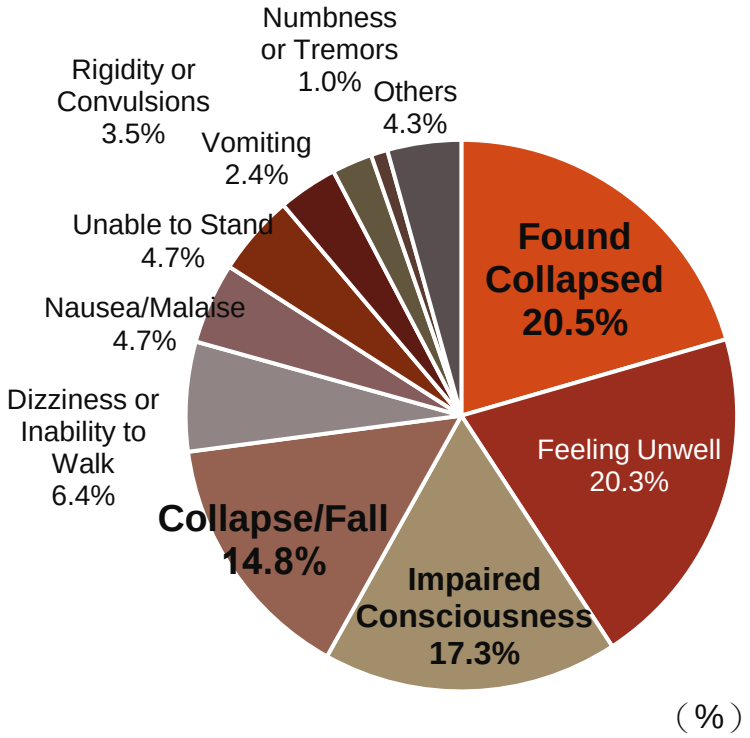
(N=372)

Amended Ordinance on Industrial Safety and Health

(Promulgated: April 15, 2025 / Enforced: June 1, 2025)

Background

- Increasing occupational heat illness cases and fatalities
- Many deaths associated with delayed recognition and response to symptoms



Applicability

Work conducted:

- **WBGT ≥ 28° C** or **Air temperature ≥ 31° C**
- For **≥ 1 continuous hour** or **> 4 h/day**

No exclusions based on industry, scale, time, or whether the setting is indoors or outdoors

Amended Ordinance on Industrial Safety and Health

(Promulgated: April 15, 2025 / Enforced: June 1, 2025)

Key Requirements for Employers

Worker with suspected heat illness

↓

1. REPORTING SYSTEM

- Immediate reporting of symptoms
- Reporting by coworkers who notice signs

↓

2. RESPONSE PROCEDURE

- Remove from work
- Cool the body
- Medical evaluation / transport

↓

3. COMMUNICATION

- Inform all workers
- Emergency contacts available







PREVENTION AT WORKPLACES

-OUR STUDIES-

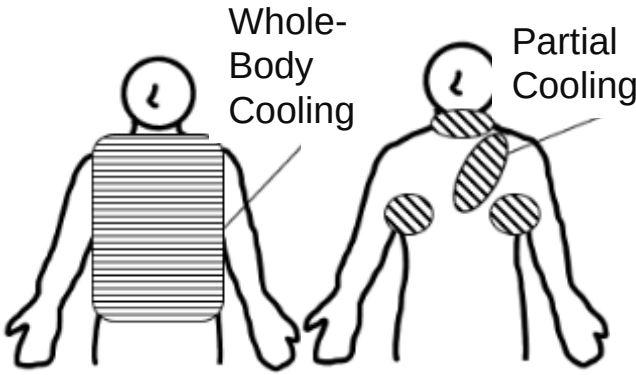
Body Cooling Methods and Their Characteristics (Japan Sports Association)

	Cooling Methods	Cooling Efficiency		Practicality				Ease of Use	Performance	Remarks
		Core	Skin	Before	Exercise	Rest	After			
External Cooling	Ice Bath	◎	◎	○	—	△	◎	△	○	May negatively affect sprint performance and muscle function immediately after cooling.
	Ice Packs	△	◎	△	△	◎	◎	◎	△	Cooling efficiency is approximately one-tenth that of an ice bath.
	Cooling Vest	△	◎	◎	○	◎	◎	◎	◎	Can be worn during exercise, although its weight may be a concern.
	Fan Cooling (Airflow/Cooling Fan)	△	○	△	—	◎	○	○	△	Can be combined with mist spraying or water spraying and can also be used outdoors.
	Head and Neck Cooling	△	◎	◎	○	◎	◎	◎	◎	Can be used during exercise, but caution is needed because core body temperature may not be sufficiently reduced.
	Palm Cooling	△	○	◎	—	◎	○	◎	○	Improves thermal sensation and can be applied in various sports.
Internal Cooling	Fluid Intake (Hydration)	○	△	◎	◎	◎	◎	◎	○	Helps prevent dehydration and provides energy supplementation.
	Ice Slurry Ingestion	◎	△	◎	△	◎	◎	◎	◎	Combines a cooling effect with electrolyte and carbohydrate replacement.

Water-Circulating Cooling Vest (Chiller System)



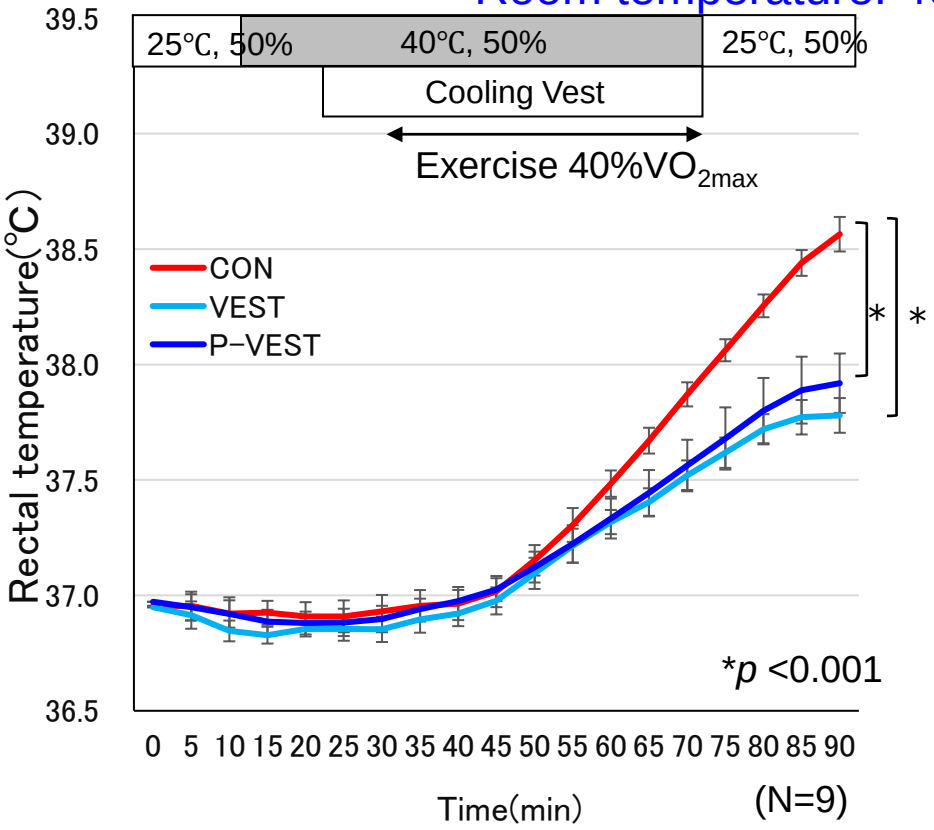


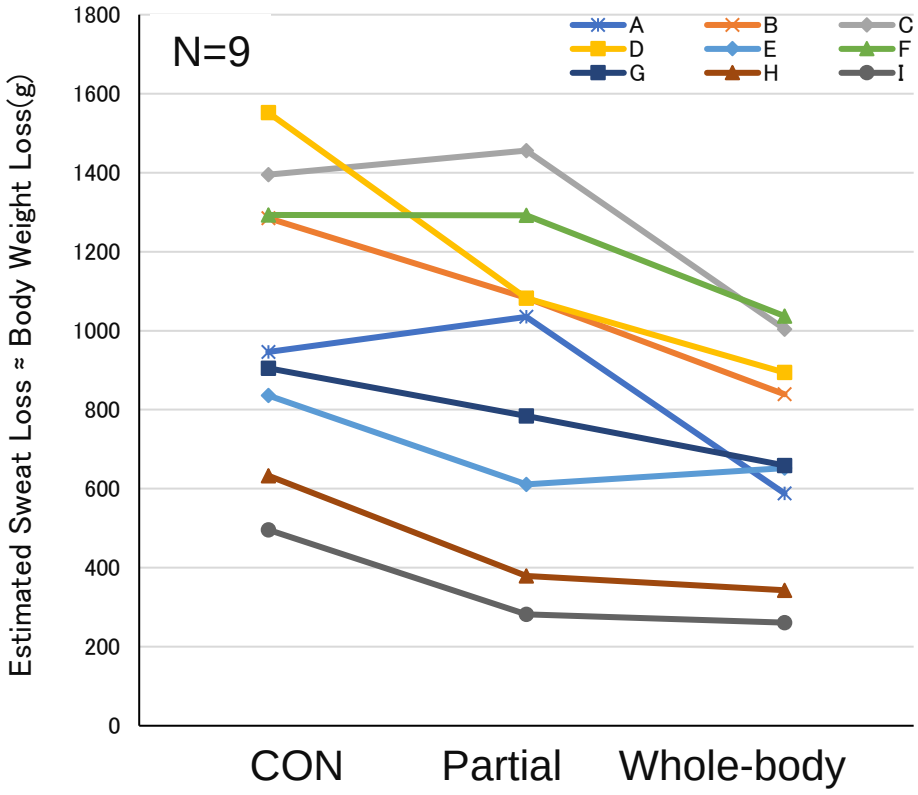


CW-WVNCLF（Kamakura Works Co., Ltd.）

Water Temperature: 10° C

Room temperature: 40° C, Relative humidity: 50%



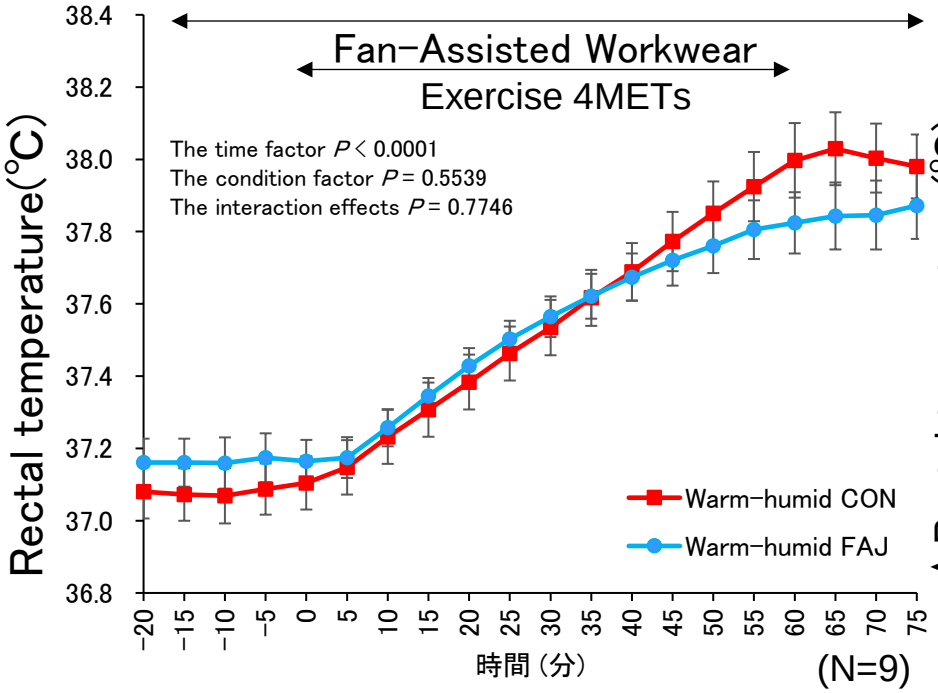


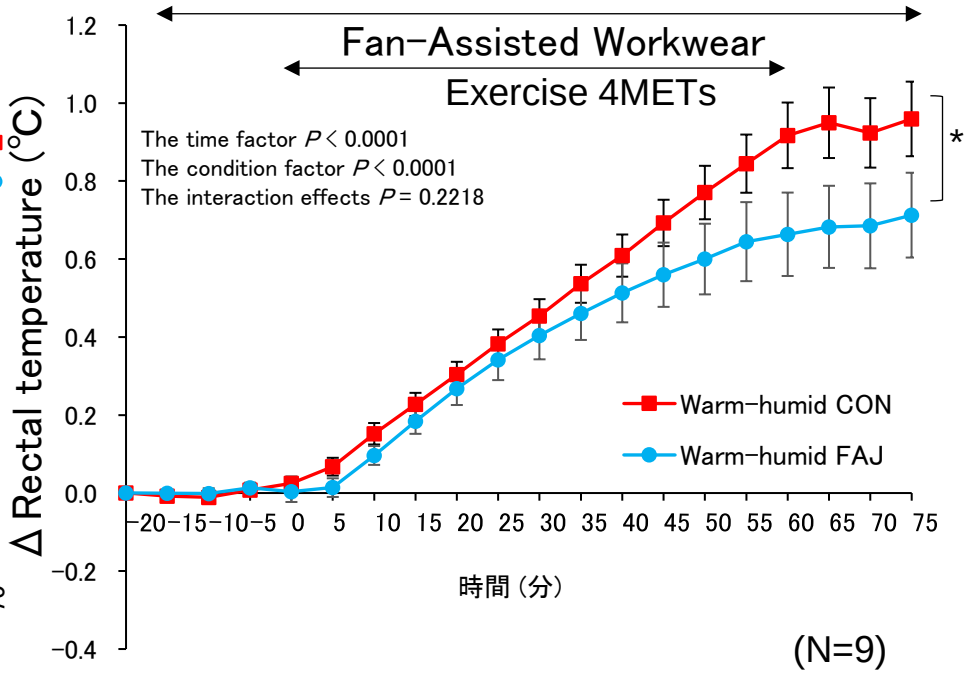
Suppression of Core Body Temperature Elevation by Fan-Assisted Workwear

- Participants: Nine healthy Japanese adult males (24.1 ± 4.3 years)
- Study Period: September 2019 to March 2020
- Study Design: Crossover study in which each participant completed four conditions (2 environmental conditions × fan-assisted workwear on/off)
- Fan-Assisted Workwear: Model KU90720, manufactured by Kuchofuku Co., Ltd.



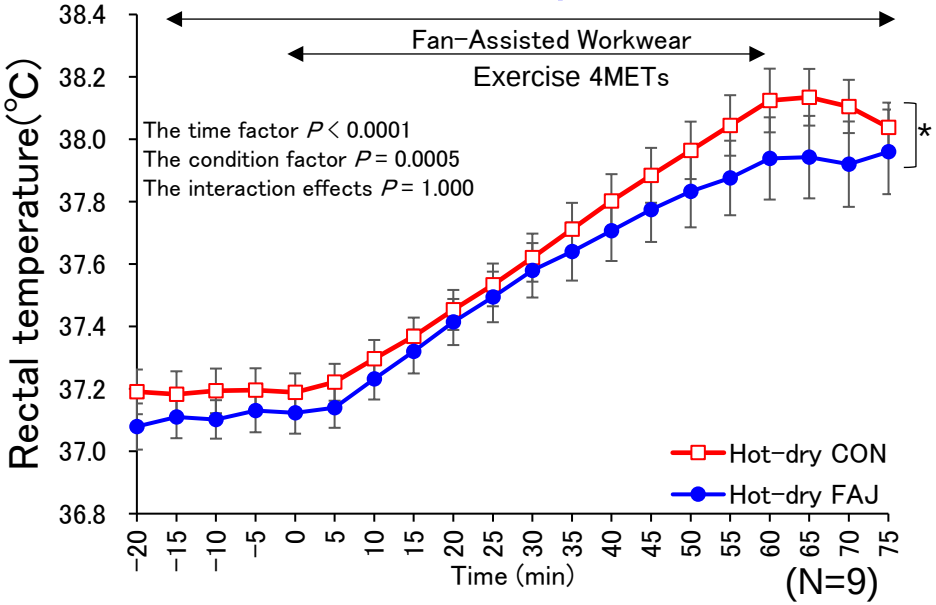
Room temperature: 30° C, Relative humidity: 85%

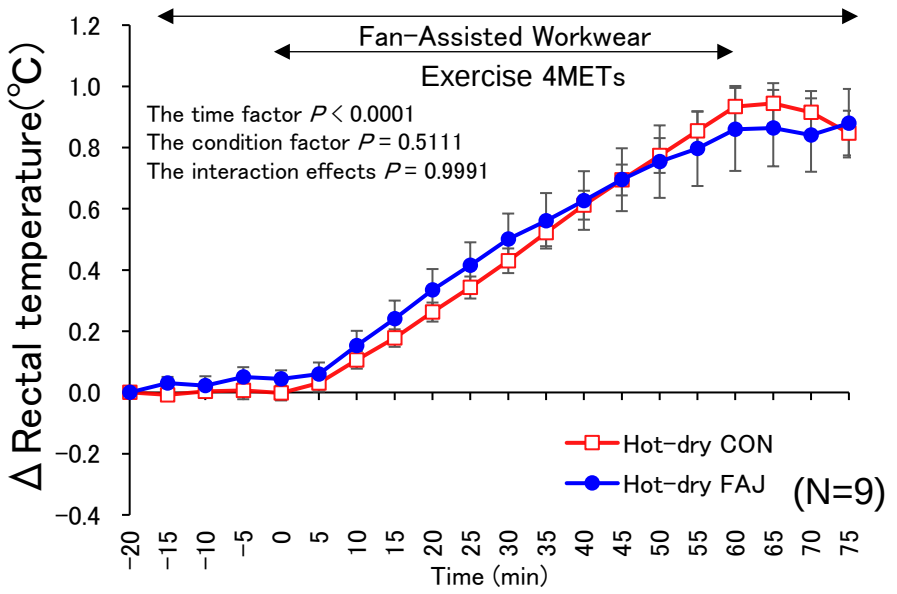


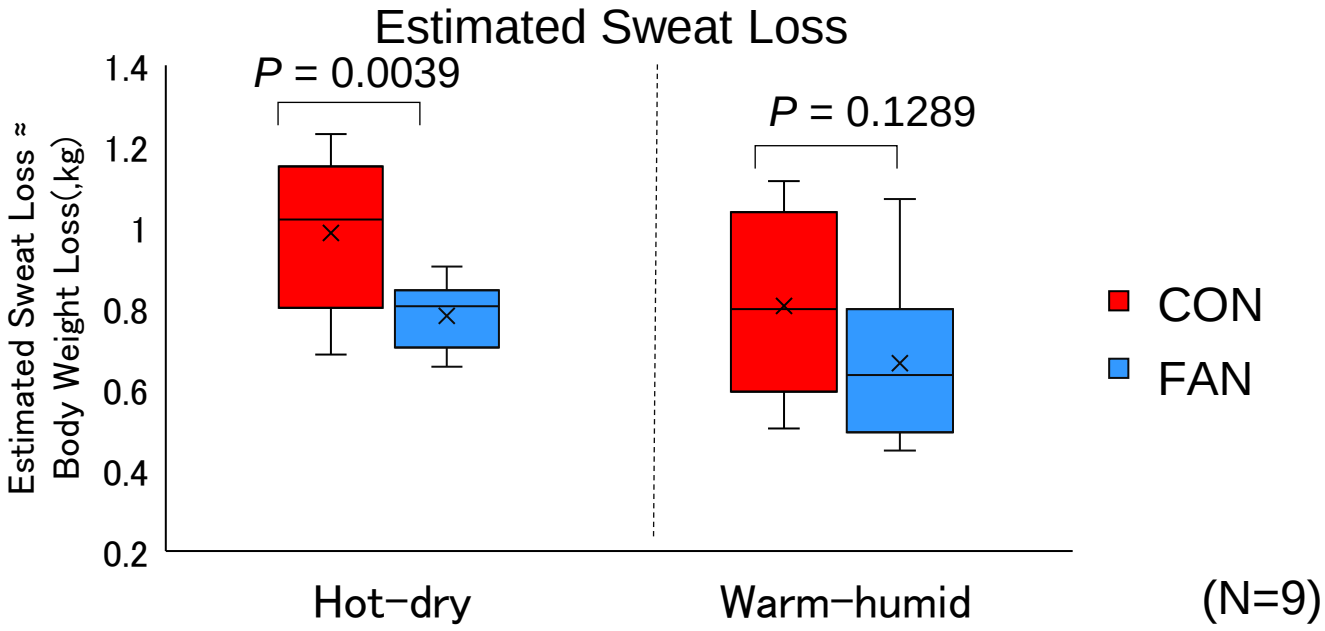


Mori K, Nagano C, Fukuzawa K, et al. Mitigation of heat strain by wearing a long-sleeve fan-attached jacket in a hot or humid environment. J Occup Health. 2022;64:e12323.

Room temperature: 40° C, Relative humidity: 30%





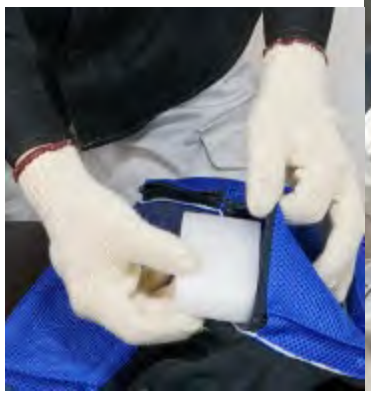




Mori K, Nagano C, Fukuzawa K, et al. Mitigation of heat strain by wearing a long-sleeve fan-attached jacket in a hot or humid environment. J Occup Health. 2022;64:e12323.

Suppression of Core Body Temperature Elevation by a Dry Ice Cooling Jacket

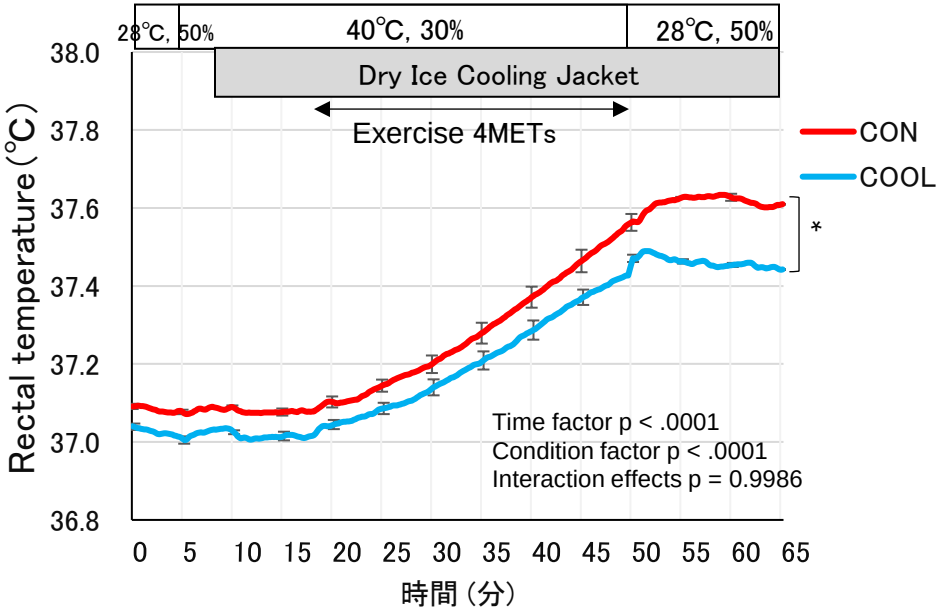
- Participants: 14 healthy Japanese adult males
- Study period: Dec 2024-Apr 2025
- Dry ice supplied via an on-demand delivery service or an on-site dry ice maker rental system
- ENEOS Dry Ice Jacket (ENEOS Amenity Co., Ltd.)

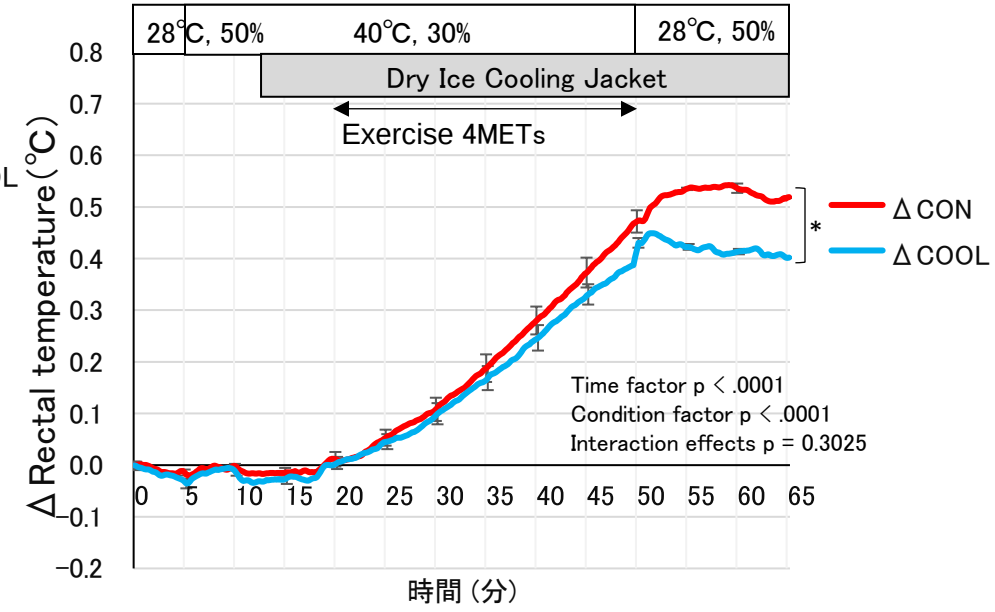






Room temperature: 40° C, Relative humidity: 30%





Effect of a Chiller-Based Seat Cooler on Suppressing Core Body Temperature Elevation

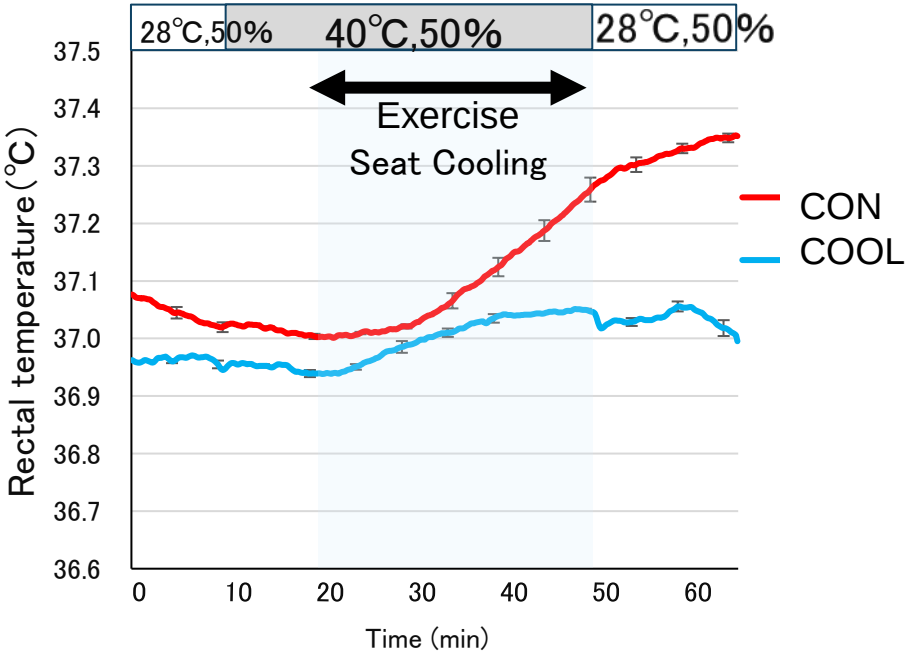
- Participants: 17 healthy Japanese adult males
- Study period: March-August 2024
- Chiller temperature setting: 7° C
- Exercise protocol: 30 min of upper-body ergometer exercise at 2.5 W

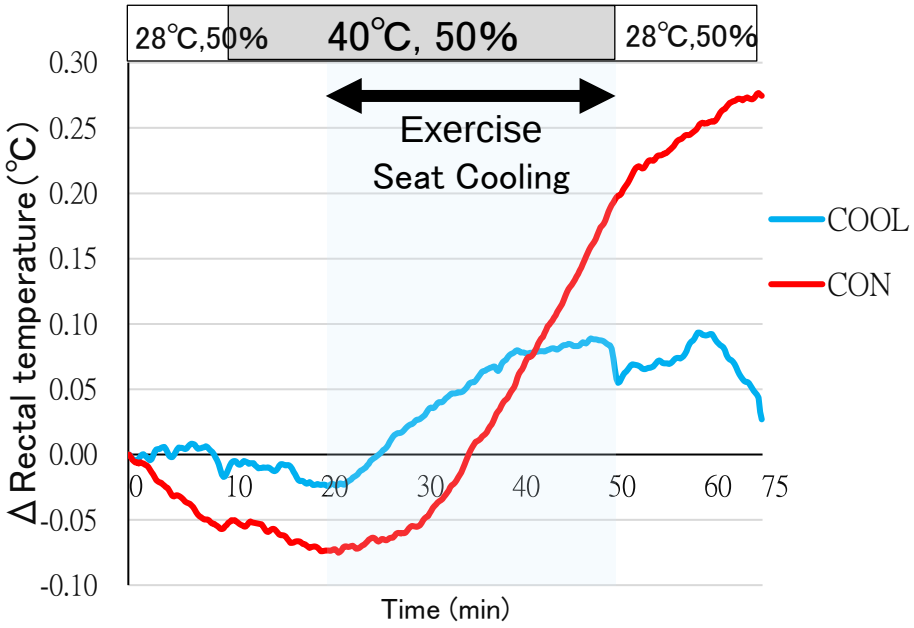


COOLEX-V152 (Kamakura Works Co., Ltd.)



Room temperature: 40° C, Relative humidity: 50%

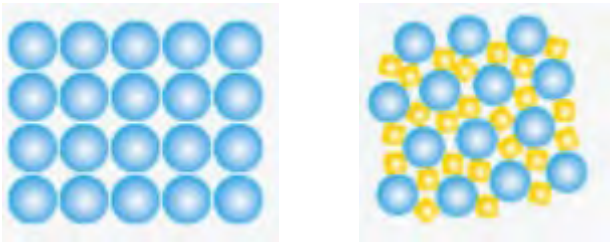




Ice Slurry for Heat Illness Prevention

What is Ice Slurry?

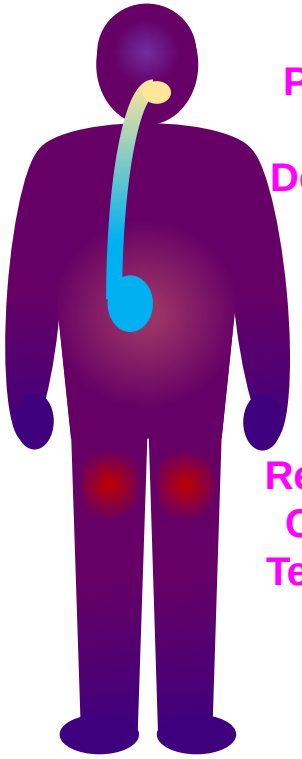
- A semi-frozen beverage containing fine ice particles suspended in liquid
- Provides simultaneous **cooling** and **hydration**
- Can be supplemented with **electrolytes** and **carbohydrates**



Ice Slurry Products



Ice Slurry Ingestion



Prevention of Dehydration

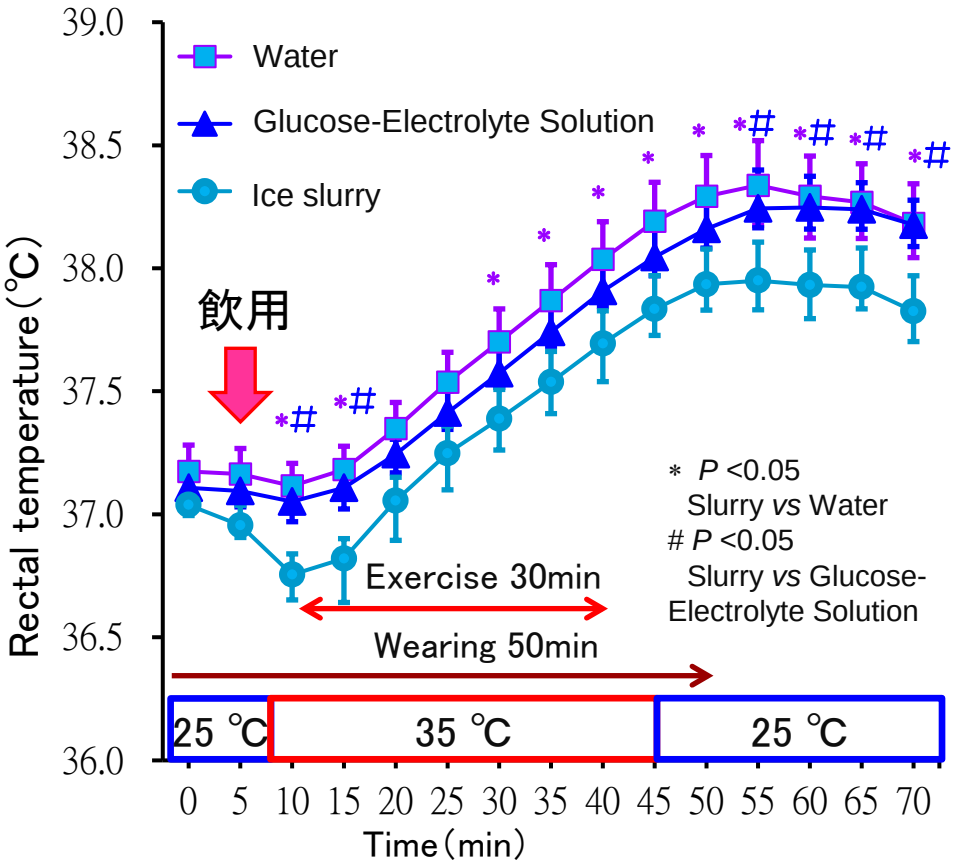
Reduction in Core Body Temperature

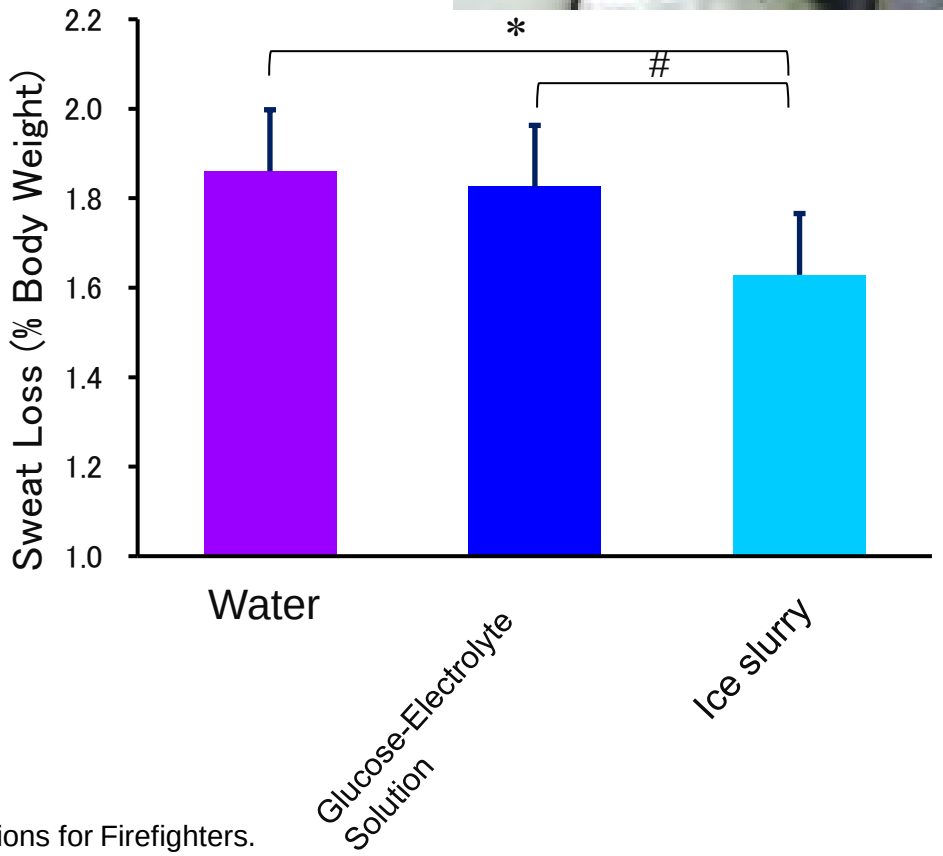
Pre-Cooling Effects of Ice Slurry

- Participants: 12 active firefighters (24.4 ± 4.3 years, male)
- Ice slurry (-1.7° C, 5.0 g/kg body weight) was compared with a carbohydrate-electrolyte solution and water



Room temperature: 40° C, Relative humidity: 50%



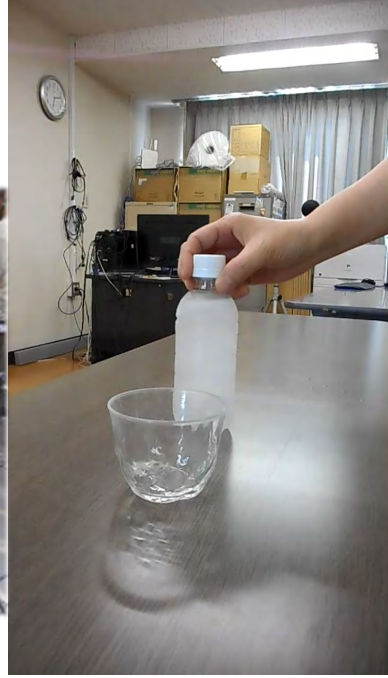


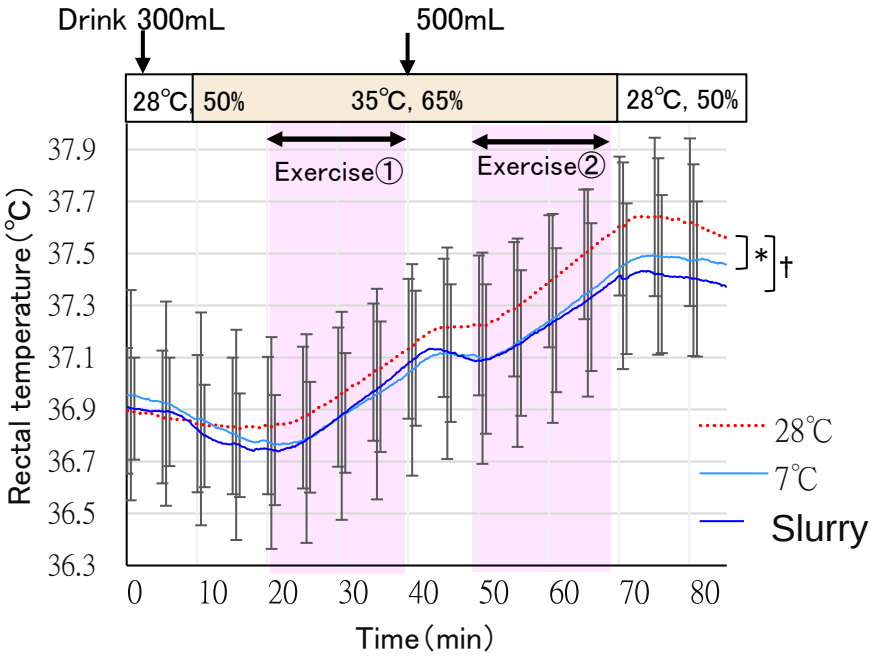
Tabuchi S, et al, Efficacy of ice slurry and carbohydrate–electrolyte solutions for Firefighters. Journal of Occupational Health. 2021;63(1) :e12263.

Effectiveness of an Ice Slurry Refrigerator

- Participants: 16 healthy adult males (23.1 ± 2.9 years)
- Study period: Dec 2024-Jun 2025
- Experimental conditions:
  - carbohydrate-electrolyte beverage (28° C / 7° C / Ice slurry)
- Beverage intake:
  - 300 mL before exercise
  - 500 mL during the inter-exercise rest period
- Ice slurry refrigerator: developmental prototype (Sharp Corporation)





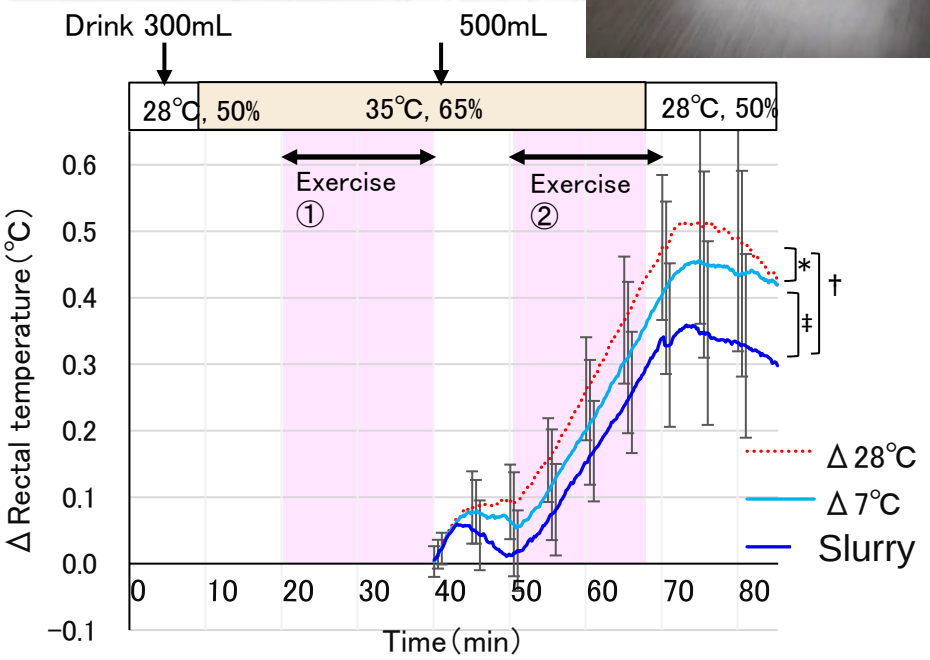


Mixed Model

Condition factor	$P < 0.0001$
Time factor	$P < 0.0001$
Interaction effects	$P = 0.9871$

Multiple comparisons

Slurry vs 28℃	$P < 0.0001$ †
7℃ vs 28℃	$P < 0.0001$ *
Slurry vs 7℃	$P = 0.5601$ ‡



Mixed Model

Condition factor	$P < 0.0001$
Time factor	$P < 0.0001$
Interaction effects	$P = 0.2329$

Multiple comparisons

Slurry vs 28℃	$P < 0.0001$ †
7℃ vs 28℃	$P < 0.0001$ *
Slurry vs 7℃	$P < 0.0001$ ‡

Challenges for Heat Illness Prevention in Japan

- Although a wide variety of heat illness prevention products are available, it is essential to select appropriate countermeasures based on an understanding of **human thermoregulatory mechanisms and the pathophysiology of heat illness,** while also considering workplace conditions and workers' health status.
- Currently, there is **no standardized certification or performance standard for heat illness prevention products.**

  The development of such standards should be considered in the future.

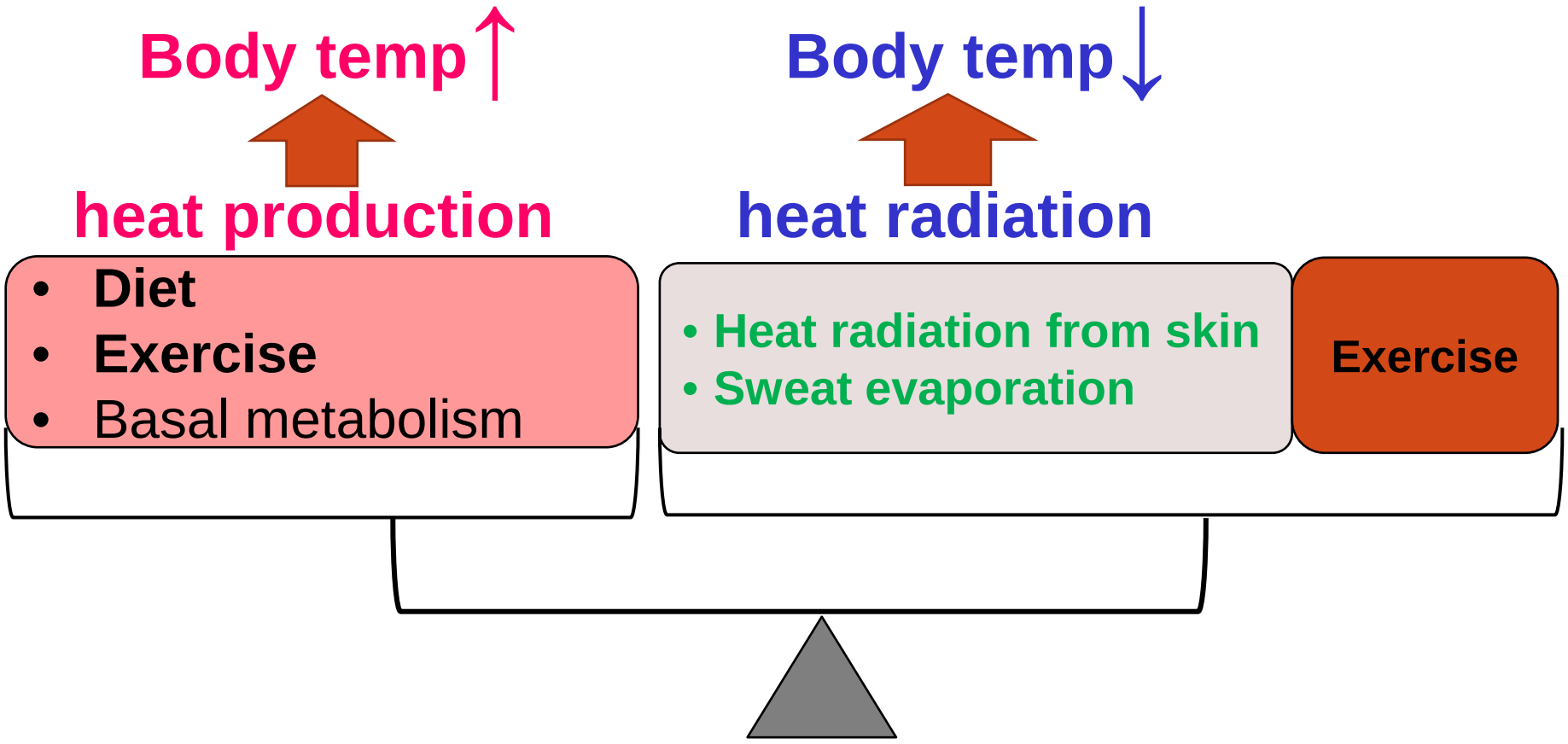
Thank you very much for your attention!
謝謝大家！



THERMOREGULATION AND TYPE OF HEAT STROKE



Balance of thermoregulation

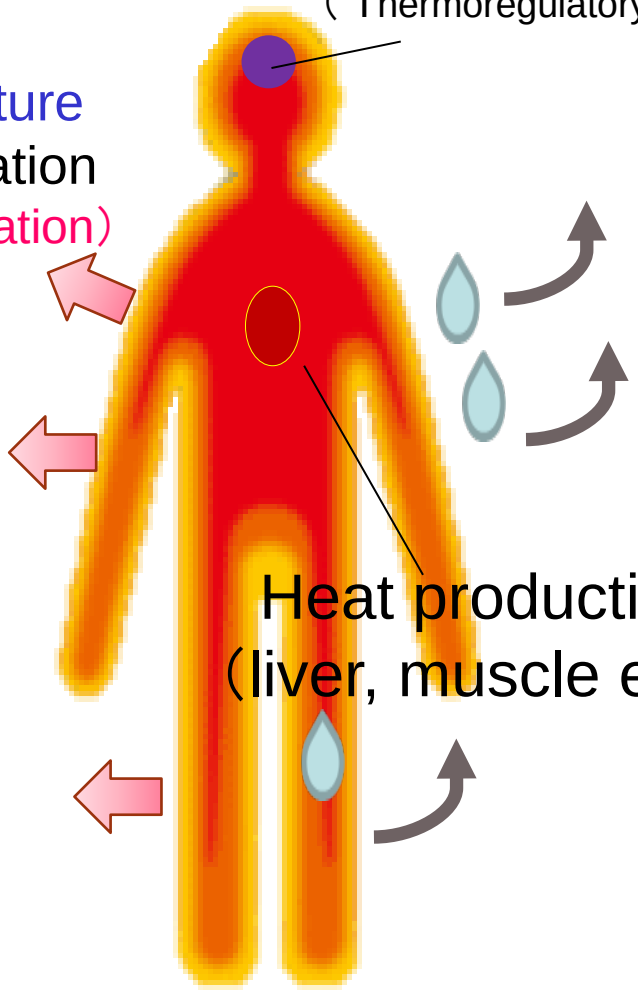


Mechanism of heat radiation

Hypothalamus
（Thermoregulatory center）

Ta＜body temperature
Cutaneous vasodilation
（Non-evaporative radiation）

Ta＞body temperature
perspiration
（Evaporative radiation）

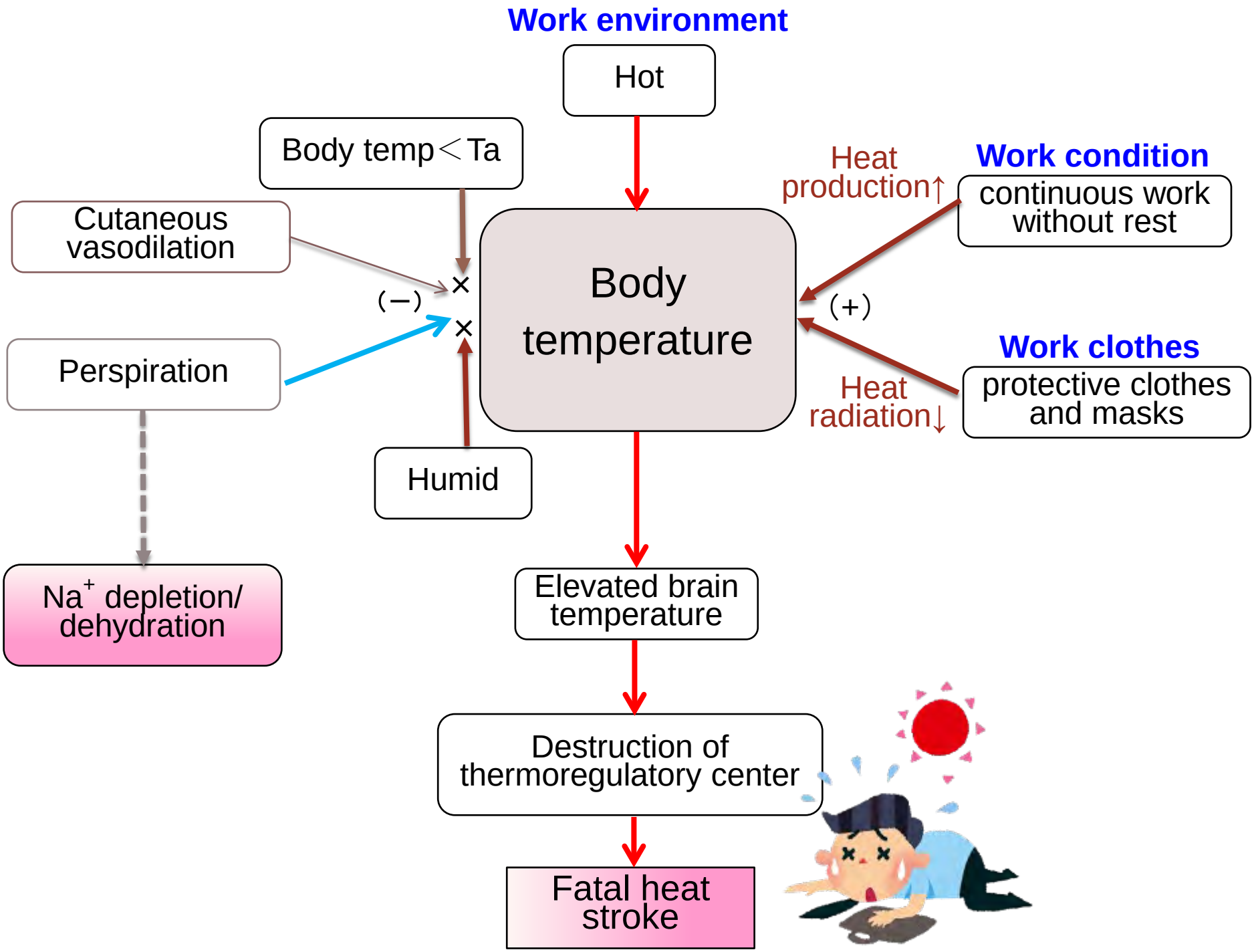


Evaporation↑↑

Heat production
（liver, muscle etc.）

Dehydration

*Ta : Air temperature


Work environment
Hot
Body temp＜Ta
Cutaneous vasodilation
Perspiration
Humid
(−)
Body temperature
Heat production↑
Work condition
continuous work without rest
(+)
Work clothes
protective clothes and masks
Heat radiation↓
Na+ depletion/ dehydration
Elevated brain temperature
Destruction of thermoregulatory center
Fatal heat stroke